

Toward Realization of Low-Altitude Economy Networks: Core Architecture, Integrated Technologies, and Future Directions

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Abstract—The rise of the low-altitude economy (LAE) is propelling urban development and emerging industries by integrating advanced technologies to enhance efficiency, safety, and sustainability in low-altitude operations. The widespread adoption of uncrewed aerial vehicles (UAVs) and electric vertical takeoff and landing (eVTOL) aircraft plays a crucial role in enabling key applications within LAE, such as urban logistics, emergency rescue, and aerial mobility. However, unlike traditional UAV networks, LAE networks encounter increased airspace management demands due to dense flying nodes and potential interference with ground communication systems.

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In addition, there are heightened and extended security risks in real-time operations, particularly the vulnerability of low-altitude aircraft to cyberattacks from ground-based threats. To address these, this paper first explores related standards and core architecture that support the development of LAE networks. Subsequently, we highlight the integration of technologies such as communication, sensing, computing, positioning, navigation, surveillance, flight control, and airspace management. We also analyze how generative artificial intelligence (GAI) contributes to this integration by enabling intelligent adaptation, uncertainty modeling, and real-time optimization. This synergy of multi-technology drives the advancement of real-world LAE applications, particularly in improving operational efficiency, optimizing airspace usage, and ensuring safety. Finally, we outline future research directions for LAE networks, such as intelligent and adaptive optimization, security and privacy protection, sustainable energy and power management, quantum-driven coordination, generative governance, and three-dimensional (3D) airspace coverage, which collectively underscore the potential of collaborative technologies to advance LAE networks.

Index Terms—Low-altitude economy (LAE) networks, uncrewed aerial vehicles (UAVs), electric vertical takeoff and landing (eVTOL), generative artificial intelligence (GAI), multi-technology integration.

I. INTRODUCTION

WITH the acceleration of globalization and urbanization, many regions are facing increasingly serious challenges, such as shortages of computational resources and inefficiencies in system performance. For example, the growing complexity of ground transportation systems leads to network congestion, reduced positioning accuracy, and overloaded computational resources, thus affecting the efficiency and responsiveness of urban operations [1]. At the same time, logistics in suburban and remote areas encounters even more complex issues, such as inconvenient transportation, high costs, and poor timeliness, which severely hamper the flow of goods and emergency response capabilities [2]. In this context, while traditional uncrewed aerial vehicle (UAV)-enabled systems have made progress [3], [4], [5], [6], [7], [8], [9], [10], they still face significant barriers in regulation, technology, and scalability [11]. These systems struggle to effectively meet the multi-layered demands of cross-regional and complex environments, and they are unable to coordinate

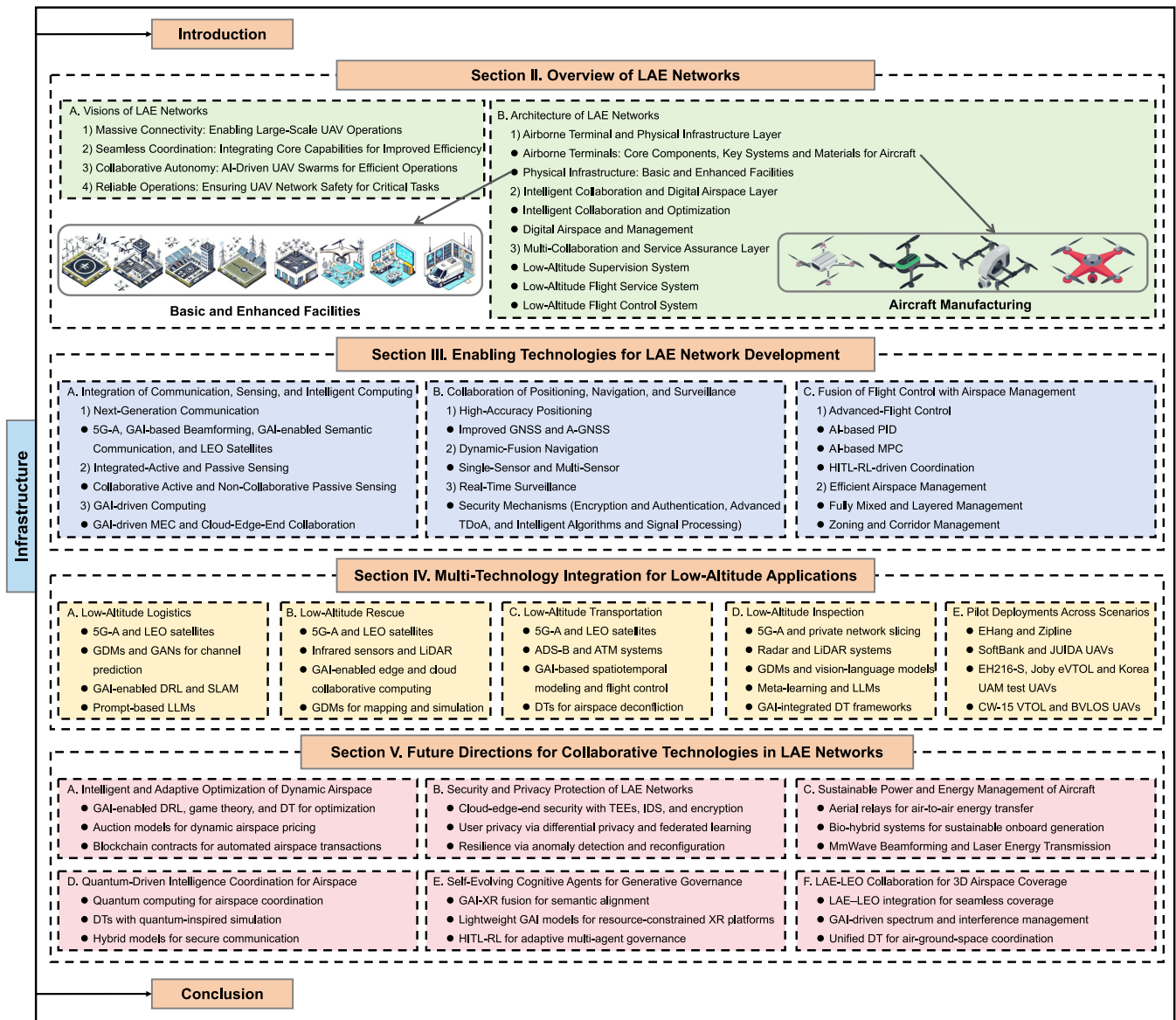


Fig. 1. The survey is structured as follows: Overview of LAE Networks (Section II), Enabling Technologies for LAE Network Development (Section III), Multi-Technology Integration for Low-Altitude Applications (Section IV), and Future Directions for Collaborative Technologies in LAE Networks (Section V).

and provide sustainable services in high-density airspace and resource-constrained areas. Therefore, there is an urgent need for an innovative, green, and intelligent solution that can overcome these limitations. The solution can facilitate more efficient, flexible, and sustainable technologies to drive long-term development across various sectors.

As an emerging industrial form, the low-altitude economy (LAE) provides a highly effective solution to current challenges by leveraging their significant potential advantages. Specifically, by unlocking the underutilized strategic airspace between 500 and 3000 meters above the ground and integrating dynamic management with layered operational models, LAE significantly enhances airspace efficiency and flexibility. Moreover, the electric vertical take-off and landing (eVTOL) aircraft [12] represent a rapidly advancing class of aerial vehicles that utilize electric propulsion systems for vertical take-off, hovering, and landing. Unlike conventional

aircraft, they typically employ distributed electric propulsion systems, which eliminate the need for runways and offer advantages such as low noise, simplified structures, and system redundancy. These characteristics make them especially suitable for operations in low-altitude airspace. Furthermore, breakthroughs in hydrogen-powered propulsion systems [13] and distributed thrust technologies [14] support aircraft with extended range and reduced dependence on traditional fuel sources. In addition, using shared airspace (e.g., the layered operation of logistics aircraft alongside crewed aircraft) and reusable infrastructure (e.g., rooftop landing platforms that double as logistics hubs), the LAE ensures efficient use of resources. Furthermore, generative artificial intelligence (GAI) technology has been applied to the optimization of communication and beamforming in LAE networks, which demonstrates the potential of GAI to enhance operational performance and system robustness [15], [16]. Given these

TABLE I
LIST OF ACRONYMS

Acronym	Definition	Acronym	Definition
ADS-B	Automatic Dependent Surveillance-Broadcast	AI	Artificial Intelligence
AR	Augmented Reality	ATM	Air Traffic Management
DDPG	Deep Deterministic Policy Gradient	DRL	Deep Reinforcement Learning
DT	Digital Twin	DQN	Deep Q-Network
eVTOL	Electric Vertical Take-Off and Landing	GAIL	Generative Adversarial Imitation Learning
GAI	Generative Artificial Intelligence	GAN	Generative Adversarial Network
GDM	Generative Diffusion Model	GNSS	Global Navigation Satellite System
GNN	Graph Neural Network	HITL-RL	Human-in-the-Loop Reinforcement Learning
ICL	In-Context Learning	INS	Inertial Navigation System
IoT	Internet of Things	ISAC	Integrated Sensing and Communication
ISCC	Integrated Sensing, Communication, and Computation	ISC3	Integration of Sensing, Communication, Computation, and Control
LAE	Low-Altitude Economy	LIDAR	Light Detection and Ranging
LLM	Large Language Model	LSTM	Long Short-Term Memory
MEC	Mobile Edge Computing	ML	Machine Learning
MIMO	Multiple-Input Multiple-Output	MR	Mixed Reality
NOMA	Non-Orthogonal Multiple Access	PPO	Proximal Policy Optimization
RAG	Retrieval-Augmented Generation	RIS	Reconfigurable Intelligent Surface
SLAM	Simultaneous Localization and Mapping	TD3	Twin Delayed Deep Deterministic Policy Gradient
UAV	Uncrewed Aerial Vehicle	VAE	Variational Autoencoder
VR	Virtual Reality	XR	Extended Reality

advantages, LAE demonstrates broad application potential in various sectors, such as logistics, rescue, transportation, and inspection. In addition to fostering diverse applications, LAE also drives innovation in UAVs [17], [18], [19], artificial intelligence (AI) [20], [21], and other technologies [22], [23], [24], [25]. This will further promote the upgrade of high-tech industries and lay the foundation for the digital transformation of the economy [26].

However, fully unleashing the potential of LAE and enabling its widespread applications require the development of efficient LAE networks, which must meet several key requirements and overcome practical challenges.

Key Requirements: These are essential to enable stable and efficient low-altitude operations.

- Intelligent autonomous decision-making capabilities are crucial [27]. This can be achieved through the deep integration of communication, sensing, and computing, which create a real-time environmental sensing and adaptive response system, thereby allowing aircraft to autonomously avoid risks and optimize flight paths in complex airspace.
- A high-precision collaborative technology system is fundamental for efficient operations [28]. This requires the collaboration of multi-modal sensors, anti-jamming navigation, and real-time aircraft monitoring to ensure centimeter-level positioning accuracy, millisecond-level response times, and safe operations in dense airspace.
- An elastic resource scheduling mechanism is key to optimizing operational efficiency [29]. By leveraging dynamic data analysis and digital twin (DT) modeling, an intelligent scheduling system can respond to changes in logistics demand or weather conditions, further enhancing airspace utilization and energy efficiency.
- Effective dynamic airspace management is essential [30]. This involves developing an intelligent air traffic management system that supports the collaborative operation of heterogeneous aircraft. The system should incorporate dynamic airspace allocation algorithms and conflict

prediction models for seamless coordination between crewed aircraft, logistics UAVs, and ground transportation, while also maintaining compatibility with international airspace management standards.

Practical Challenges: Several issues still hinder large-scale deployment and coordination.

- Fragmented regulatory systems, inconsistent approval procedures, and the lack of standardized low-altitude traffic rules may hinder the coordination of airspace, communication, and service layers, thereby limiting cross-regional operations of low-altitude aircraft. Moreover, ambiguity in data privacy and compliance governance for aerial operations further impedes large-scale deployment.
- Managing spectrum usage and mitigating interference is particularly challenging due to the shared or unlicensed frequency bands used by low-altitude aircraft, edge devices, and ground stations, which is further complicated by the high mobility of aerial platforms.
- Dense urban environments introduce severe signal degradation caused by obstructions, multipath effects, and frequent non-line-of-sight (NLOS) conditions, which can compromise communication, localization, and sensing, especially in global navigation satellite system (GNSS)-denied scenarios.
- Environmental uncertainties such as rain, fog, snow, and turbulence can affect aircraft stability, communication link quality, and task execution efficiency. Unlike fixed ground infrastructure, low-altitude aircraft are highly exposed to atmospheric disturbances, imposing stricter demands on control and perception systems.

Based on the above challenges, this survey focuses on the multidimensional synergy of supporting technologies and the key challenges they face. These include the integration of communication, sensing, and intelligent computing, collaboration of positioning, navigation, and surveillance, and fusion of flight control with airspace management. Moreover, GAI plays a vital role in this synergistic process in terms of intelligent adaptation, uncertainty modeling, and real-time optimization.

TABLE II
SUMMARY OF RELATED SURVEYS VS. OUR SURVEY

Reference	Key focus of survey	How our survey differs
[31]	Designs a low-altitude intelligent transportation system framework that focuses on technologies such as UAVs, AI, and IoT.	Our survey introduces the core architecture of LAE networks and emphasizes how these enabling technologies are implemented within such networks.
[32]	Examines how GAI enhances physical-layer communication security by improving confidentiality, authentication, and robustness.	Our survey highlights the role of GAI not only in communications, but also as a key enabler for integrated and adaptive intelligence across LAE systems.
[33]	Reviews the application of GAI in physical-layer signal processing, including channel estimation, modulation recognition, and equalization.	
[34]	Explores how GAI reconstructs semantic meaning, thereby enabling generative semantic communication and avoiding bit-level recovery.	We embed semantic communication as a technical module within the LAE networks and analyze its interactions with sensing, navigation, and control.
[35]	Investigates GAI-based architectures for semantic transmission, knowledge modeling, and adaptive resource allocation.	
[36]	Provides a comprehensive review of AI-integrated ISAC to enhance adaptability and minimize model dependency.	We also incorporate multi-aircraft coordination, airspace allocation, and intelligent flight control, which are essential for stable operation in LAE networks.
[37]	Investigates the role of GAI in swarm intelligence for enabling collaborative perception and decision-making in uncrewed vehicle networks.	
[27]	Presents how EAI can be utilized to achieve the integration of ISC3 in LAE networks, thereby improving system-level intelligence and autonomy.	We further explore the critical roles of navigation, positioning, surveillance, flight control, and airspace management technologies, while examining how these integrated technologies work together to enhance the performance and security of LAE networks.
[38]	Proposes the integrated ISCC framework to support multi-resource orchestration and task-aware collaboration in future edge-intelligent networks.	

A. Related Works and Contributions

Given the popularity of LAE, several surveys have established a solid foundation for understanding the related technologies and applications. Table II summarizes these surveys and shows the distinct contributions of our own survey.

In [31], a low-altitude intelligent transportation (LAIT) system framework based on a cyber-physical system (CPS) architecture was proposed, along with a lifecycle management framework to improve the efficiency of low-altitude traffic operations. Moreover, the authors emphasized innovations in key technologies such as UAVs, AI, internet of things (IoT), 6G, cloud computing, and blockchain, which demonstrate the broad technological basis supporting LAE networks. Meanwhile, several surveys have explored the emerging role of GAI in enhancing specific domains. In [32], the authors surveyed the role of GAI in securing physical-layer communications, including confidentiality, authentication, and robustness. In addition, [33] offered a comprehensive review of GAI for physical-layer signal processing, highlighting its advantages in tasks such as channel estimation, modulation recognition, and equalization. Furthermore, [34] investigated a GAI-driven semantic communication framework that underscores information regeneration over traditional decoding, while [35] examined semantic transmission, knowledge modeling, and resource management. Despite these valuable contributions, these surveys focus on individual technological aspects and do not delve into how these technologies can be seamlessly integrated and synergized, which is critical for the comprehensive development of LAE networks.

In contrast to the above efforts, a new wave of research has emerged that integrates GAI across multiple submodules to support LAE-level coordination. For example, [36] provided an extensive survey on intelligent integrated sensing and communication (ISAC), outlining how AI incorporation enhances system adaptability and reduces reliance on explicit modeling.

Moreover, [37] addressed the challenges of swarm intelligence in uncrewed vehicle networks and discussed how GAI can facilitate collaborative perception and decision-making among multiple agents. Furthermore, [27] explored how embodied artificial intelligence (EAI) can be utilized to achieve the integration of sensing, communication, computation, and control (ISC3) for LAE scenarios. Additionally, [38] proposed the integrated sensing, communication, and computation (ISCC) framework, which provides a holistic solution for multi-resource orchestration and task-aware collaboration in future edge-intelligent networks. However, the aforementioned surveys predominantly emphasize communication, sensing, and computing technologies, while underestimating the essential roles of navigation, positioning, surveillance, flight control, and airspace management.

Different from existing surveys and tutorials, our survey distinguishes itself by focusing specifically on the core technological architecture and applications of LAE networks. Moreover, this survey offers a unique perspective by deeply integrating communication, sensing, intelligent computing, positioning, navigation, flight control, and airspace management technologies. In particular, we highlight the enabling role of GAI in enhancing system coordination and adaptability under dynamic conditions. Therefore, this survey fills a gap in the literature by thoroughly exploring how the synergy of these technologies enables efficient and secure operations in LAE networks, and by emphasizing their critical role in addressing challenges such as aircraft coordination and airspace management in complex airspace scenarios.

The main contributions of this paper are summarized as follows:

- We highlight the distinctive features of LAE networks compared to traditional UAV networks and describe the multi-layer architecture that supports their development. This establishes a technical framework for understanding and evaluating LAE implementations.

TABLE III
GLOSSARY OF KEY TERMS

Term	Explanation	Term	Explanation
LAE Networks	Networked infrastructures that support large-scale, intelligent, and coordinated operations of low-altitude aircraft	UAV Swarms	Multiple UAVs that collaborate autonomously to complete coordinated tasks
Digital Airspace	A virtualized and programmable low-altitude environment for dynamic flight scheduling, routing, and airspace monitoring	GAI-driven MEC	MEC architecture enhanced by GAI for real-time decision-making and resource adaptation
GAI-driven Cloud-Edge-End Collaboration	Leveraging GAI to enable seamless coordination, adaptive interaction, and content generation across cloud, edge, and end nodes	High-Accuracy Positioning	Fine-grained localization of low-altitude aircraft using GNSS, INS, and sensor fusion
Dynamic Fusion Navigation	Real-time navigation that adaptively integrates multiple data sources such as LiDAR, SLAM, and GNSS to improve robustness and path accuracy	Real-Time Surveillance	Continuous, low-latency tracking of aircraft status and airspace environment to ensure safety and compliance
Advanced Flight Control	Intelligent and adaptive control systems that ensure stable, accurate, and robust operations of aircraft in complex environments	Efficient Airspace Management	Dynamic coordination and optimization of airspace to ensure safe, conflict-free, and high-capacity operations across diverse scenarios

- We explore how the deep integration of communication, sensing, intelligent computing, positioning, navigation, surveillance, flight control, and airspace management technologies drives the efficient and safe operation of LAE networks. We also analyze how GAI contributes to this integration by enabling intelligent adaptation, uncertainty modeling, and real-time optimization. These capabilities support emerging applications such as logistics, rescue, transportation, and inspection.
- We outline future research directions for LAE networks, including intelligent and adaptive optimization, security and privacy protection, sustainable energy and power management, quantum-driven coordination, generative governance, and three-dimensional (3D) airspace coverage. These discussions provide insights into how collaborative technologies can meet future operational needs in LAE systems.

The structure of this survey is outlined in Fig. 1. Section II presents the vision for LAE networks and its core architecture. Section III provides an overview of the core technologies supporting LAE networks. Section IV explores the multi-technology integration in real-world applications. Section V discusses the future research directions, and Section VI concludes the paper. To aid the reader, Table I provides a list of acronyms for quick reference, and Table III presents a glossary of key terms to enhance conceptual clarity.

II. OVERVIEW OF LAE NETWORKS

In this section, we begin by outlining the visions for LAE networks. Subsequently, we present the core architecture that supports the development and operations of LAE networks.

A. Visions Toward LAE Networks

LAE networks have driven significant advancements in UAV networks, which impose increasing demands on scalability, efficiency, and reliability. By integrating multifunctional systems and enabling autonomous collaboration, LAE networks achieve highly efficient operations. The following outlines the key aspects of the developmental vision for LAE networks and the standards that support them. In addition, Table IV presents a comparison of relevant standards to support systematic understanding and implementation.

1) Massive Connectivity: Enabling Large-Scale UAV Operations: Traditional UAV networks are typically capable of efficiently supporting only a few dozen UAVs. When this number is exceeded, issues such as signal interference and communication disruptions often arise, which fail to meet the demands of the performance in terms of real-time, flexibility, and reliability [39]. This is primarily due to the limitations of existing communication protocols and wireless resource management in high-density networks, including frequent channel switching causing interruptions and instability in communication links, and inefficient spectrum management leading to over-utilization or under-utilization of resources [40]. However, LAE networks need to support the coordinated operation of hundreds or even thousands of UAVs, which places greater demands on communication link stability, efficient spectrum utilization, and effective route management.

To support such large-scale connectivity, the IEEE 1939.1-2021 standard [41] introduces new technologies that collectively form a structured and scalable low-altitude airspace framework. Specifically, the standard clearly defines communication link quality requirements and network security standards to ensure stable communications in dense environments, and it also optimizes the low-altitude airspace structure and flight route design through communication coverage tests in the target airspace. Moreover, the standard proposes a grid-based route planning method, which divides the airspace into $1\text{ km} \times 1\text{ km}$ grids and plans independent flight paths within each grid to support the coordinated operation of multiple UAVs. Furthermore, the standard integrates enhanced mobile broadband (eMBB) and ultra-reliable low-latency communication (URLLC), thereby guaranteeing reliable performance in high-communication-load scenarios through dynamic frequency adjustment and real-time adaptability. In addition, 3GPP Release-18 specifies new radio (NR)-UAV enhancements,¹ including UAV mobility management, network slicing, and interference mitigation, which further improve the scalability and robustness of UAV connectivity.

2) Seamless Coordination: Integrating Core Capabilities for Improved Efficiency: In traditional UAV networks, functions

¹<https://www.3gpp.org/technologies/nr-uav>

TABLE IV
COMPARATIVE OVERVIEW OF RELEVANT STANDARDS FOR UAV SYSTEMS

Standard	Scope	Technical Focus	Applicability
IEEE 1939.1-2021	Provides a structured framework for managing low-altitude UAV airspace	• Grid/corridor-based segmentation • Communication and sensing protocols • Flight planning and conflict avoidance	• Enables scalable UAV operations in urban and rural environments • Facilitates national UTM systems and urban air mobility • Serves regulators, operators, and UTM developers
IEEE 1937.8-2024	Specifies hardware and software requirements for UAV terminals	• Interfaces for power, data, and antennas • UART/Ethernet communication protocols • Security features: encryption, authentication, firmware protection • Environmental reliability: EMC, MTBF, shock, voltage tolerance	• Facilitates standardized terminal integration and interoperability • Ensures reliable operation in harsh conditions • Targets manufacturers and system integrators
IEEE 1937.3-2024	Establishes GNSS-based flight data transmission for UAVs	• 35-byte format: ID, position, speed, timestamp • GNSS short-message communication • Real-time supervision without cellular networks	• Supports operations in remote or weak-signal regions • Enables flight monitoring and regulatory compliance • Applies to UAVs equipped with GNSS modules
IEEE P1954	Defines protocols for self-organizing and spectrum-agile UAV communication	• Autonomous networking and spectrum adaptation • Compatible with IEEE P1920.1 and P1900 series • Enhanced air-to-air communication	• Enables rapid deployment in infrastructure-limited scenarios • Coordinates UAV swarms for disaster response and aerial tasks • Involves UAV manufacturers, ICT vendors, and policymakers
3GPP NR-UAV (Release 18)	Presents enhancements in 5G NR for UAV connectivity and control	• UAV positioning and mobility management • Network slicing and QoS provisioning • Interference detection and mitigation	• Supports scalable UAV operations in 5G-A networks • Enables commercial and industrial UAV applications • Strengthens UAV integration into mobile infrastructure
ITU-T Y.4421	Defines system architecture for UAVs and UAV controllers via IMT-2020 networks	• UAV-controller communication • Service orchestration, authentication, and QoS handling	• Facilitates task-oriented network-based control • Enhances coordination efficiency in dynamic multi-UAV operations
ITU-R M.2171	Highlights spectrum requirements and coexistence analysis for UAV systems	• Frequency allocation and spectrum sharing • Coexistence with existing radio services • Interference modeling and assessment	• Assists in UAV-related spectrum planning • Informs regulatory decisions for aerial frequency allocation

such as sensing, navigation, communication, and control often operate independently rather than being effectively coordinated. For example, the sensing module collects environmental data by using thermal imaging cameras, light detection and ranging (LiDAR), or gas sensors [42], which is then transmitted by the communication module to a ground control center for processing [43]. Subsequently, the navigation module adjusts the flight path based on the returned results. This multi-step interaction increases the data transmission delays and limits the real-time task performance, especially in complex and dynamic scenarios. In contrast, LAE networks require seamless integration of these functions into a unified framework to enable real-time data exchange and efficient task optimization.

To make this possible, the IEEE 1937.8-2024 standard [44] proposes the seamless integration of these key functions through the coordinated operation of cellular communication terminals and other modules. In particular, the cellular communication terminal, as the core module of the UAV, supports beyond visual line of sight (BVLOS) flight control and real-time transmission of high-definition video data. Moreover, it is also responsible for collecting, processing, and transmitting flight data (e.g., position, speed, and attitude) and the task payload data (e.g., video streams and sensor

information). By interacting with ground control centers, the terminal enables the real-time execution of flight commands, as well as precise path planning and navigation. In addition, to ensure efficient data transmission, the standard specifies the use of both serial and Ethernet interfaces, with a requirement that the switching delay of cellular base stations must not exceed 5 milliseconds to maintain continuous communication. Moreover, ITU-T Y.4421² defines a functional architecture for UAV-controller interaction, coordinated communication, task orchestration, and service management in complex network environments.

3) *Collaborative Autonomy: AI-driven UAV Swarms for Efficient Operations:* Traditional UAV networks heavily rely on human intervention, which requires ground operators to handle task planning, path adjustments, and fault management [11]. This dependency leads to a lack of autonomy in adapting to dynamic environments and unforeseen situations. Moreover, task execution is often decentralized, with UAVs lacking coordination mechanisms and information sharing [45], which result in low resource utilization and diminish the effectiveness of task completion. The LAE networks can address the above-mentioned issues by incorporating AI into the UAV systems,

²<http://www.itu.int/rec/T-REC-Y.4421-202110-I>

particularly leveraging artificial general intelligence (AGI) [46] and GAI [47] to enable UAVs to autonomously learn, plan, and collaborate [48], thereby achieving a high level of autonomy within UAV swarms [49].

To better integrate AGI and enable efficient operations of UAV swarms, the P1954 standard [50] defines an architecture and protocol that support self-organizing and spectrum-flexible communication for UAVs, which allows them to automatically establish networks, dynamically adapt spectrum resources, and maintain connections with ground users and devices. The core components include dynamic network deployment, self-organizing spectrum management functions, and integration with existing communication standards such as the IEEE P1900 series. This makes it applicable to disaster emergency communications and the coordination of high-density UAV swarms. In addition, the standard incorporates air-to-air communication specifications from IEEE P1920.1 [51], along with the dynamic spectrum access and interference co-existence analysis from the IEEE 1900 series, thereby ensuring the reliability and efficiency of communication.

4) *Reliable Operations: Ensuring UAV Network Safety for Critical Tasks:* Critical tasks such as air taxi services, emergency response, and medical supply transport place high demands on the reliability and safety of UAV networks [52], [53], [54]. However, traditional UAV networks rely on limited communication links for redundant systems, and the adaptability of multi-path technologies is often insufficient due to technical constraints [55]. As a result, UAVs are more prone to communication disruptions or data loss when facing signal interference, network congestion, or other unforeseen events, which can impact task execution and even cause task failure [56]. Therefore, to ensure the reliable operation of critical tasks in LAE networks, relevant standards provide comprehensive specifications for flight supervision data transmission and short message transmission protocols.

The IEEE 1937.3-2024 standard [57] specifies the content of UAV flight monitoring data, including information such as identity, status, latitude, longitude, altitude, heading, speed, and timestamps. Moreover, it also mandates that the data sampling interval does not exceed 2 seconds to ensure real-time monitoring of UAV flight safety, so that the accurate positioning and behavior tracking can be enabled. Furthermore, the standard utilizes the short message mechanism of the GNSS, which defines the transmission and reception protocols for flight supervision data and specifies the format for the header and data sections of the data packets. In addition, the immunity of this standard to space, territory, or terrain limitations effectively enhances the continuity and stability of flight supervision data transmission. Complementarily, ITU-R M.2171³ offers guidance on UAV spectrum requirements, coexistence strategies, and interference evaluation to support reliable and coordinated aerial operations.

B. Architecture of LAE Networks

To achieve efficient operation of low-altitude aircraft, a well-designed architecture is essential. This architecture supports

both system stability and scalability, while facilitating seamless collaboration among its components. Specifically, the details of the overall LAE network architecture are as follows. Moreover, an interaction diagram of the architecture is provided, as shown in Fig. 2.

1) *Airborne Terminal and Physical Infrastructure Layer:*

The first layer of the LAE network architecture consists of airborne terminals that are related to the aircraft manufacturing and low-altitude physical infrastructure. Specifically, the former is a crucial component of the aircraft and plays a decisive role in flight performance, safety, and reliability, and the latter focuses on building a robust physical infrastructure to support efficient low-altitude aviation operations.

Airborne Terminal: The airborne terminal provides the foundation for power support, flight control, environmental sensing, and information exchange in the aircraft, which integrates the core components and key systems that are required for the aircraft operation.

- *Core Components:* Various components perform unique functions to ensure system stability. Specifically, aviation relays manage the switching of low-power circuits for the safety of the electrical system [58]. Moreover, engine ignition systems trigger combustion, which affects engine efficiency [59]. In addition, circuit protection devices monitor and disconnect faulty circuits to prevent damage, and the aviation contactors control the high-power loads for critical equipment stability [60].
- *Key Systems:* The power system provides strong thrust for flight. For example, small and medium engines are compact and powerful for frequent takeoffs tasks. Moreover, the hybrid systems [61] offer endurance and quick response for extended flights, and the onboard systems support real-time data exchange and path planning through sensing and navigation. In addition, the flight control systems can manage the attitude and task commands to enhance autonomy and adaptability [62].

It is worth noting that the key materials for low-altitude aircraft are integral to the development of airborne terminals and aircraft manufacturing [72]. In particular, aluminum alloys, known for their high strength-to-weight ratio and strong corrosion resistance, are widely used in aircraft structures, including fixed-wing and helicopters. UAVs are ideal for using carbon fiber due to its high strength and low density. Moreover, composite materials enable the lightweight and optimized the designs for flying cars, such as eVTOLs.

Physical Infrastructure: A comprehensive and well-rounded operational support system is established to meet the diverse business needs of enterprises. Specifically, such a system includes both basic and enhanced facilities, which not only can support routine operations but also significantly improve the ability of system to respond quickly to unexpected situations and environmental changes.

- *Basic Facilities:* Takeoff and landing stations are crucial to initiating and completing flight tasks [73]. Transfer facilities enable smooth transitions between takeoff points and task areas [74]. Moreover, energy stations provide charging or battery replacement services, while

³<https://www.itu.int/pub/r-rep-m.2171>

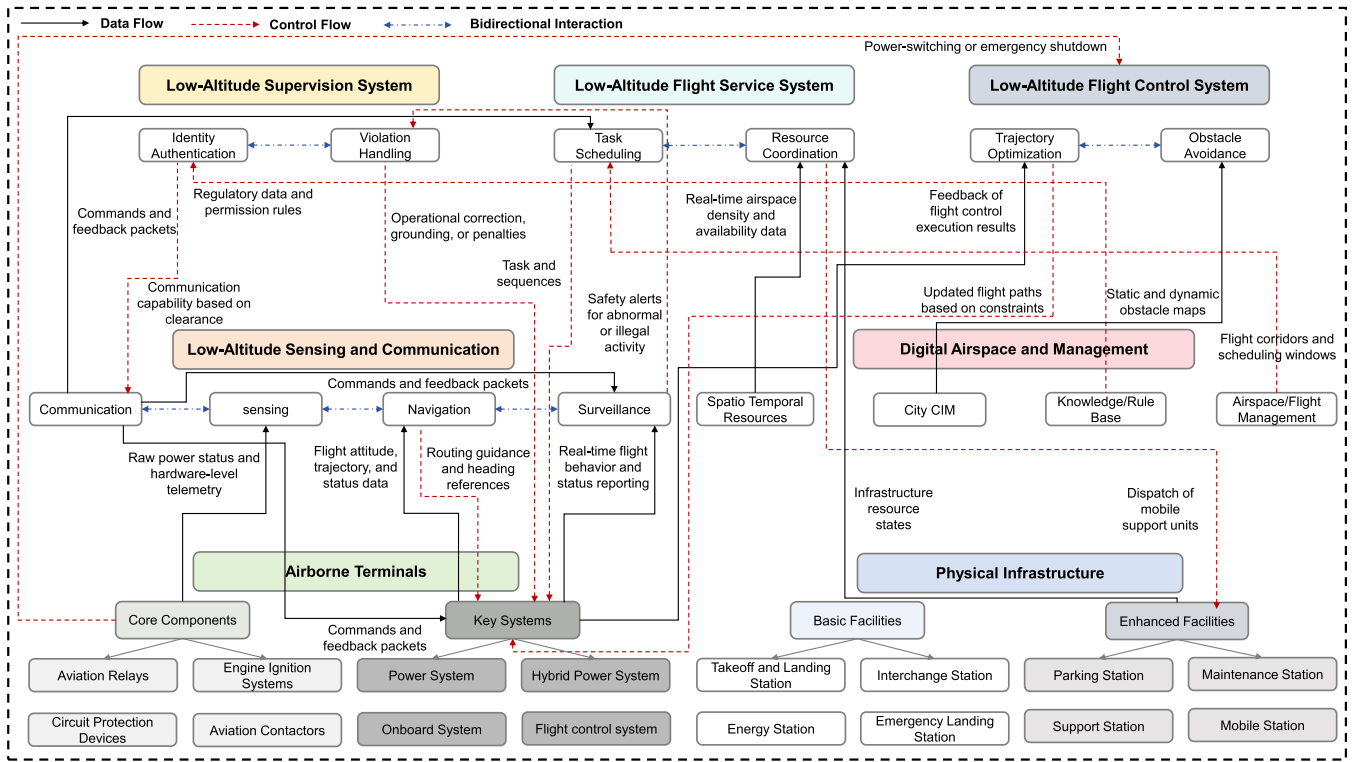


Fig. 2. Interaction diagram of the LAE network architecture. Each module performs its role by sending control commands or receiving operational data, thereby establishing a coherent network of data flow, control flow, and bidirectional interaction [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71].

emergency landing sites offer the safe options for unforeseen situations [64].

- **Enhanced Facilities:** These expand infrastructure are supported by providing secure parking for aircraft, ensuring timely maintenance, and offering support stations for task coordination. Moreover, mobile facilities (e.g., portable charging stations [63] or repair units [75]) can be quickly deployed to improve the response speed and flexibility.

2) Intelligent Collaboration and Digital Airspace Layer:

The second layer of the LAE network architecture comprises the intelligent collaboration and optimization, as well as digital airspace and management. The former ensures the real-time flow of information and environmental awareness between low-altitude aircraft and ground facilities, thereby guaranteeing the safety and efficiency of aircraft operations. The latter focuses on the optimal allocation of airspace resources and flight scheduling, which ensures the orderly use of low-altitude airspace and efficient execution of flight operations.

Intelligent Collaboration and Optimization: It is essential to transform low-altitude airspace into a computable domain, which involves various aspects such as communication, sensing, computing, positioning, navigation, and surveillance. Detailed technical aspects will be discussed in Section III.

Digital Airspace and Management: The digital airspace and management provides comprehensive digital services for low-altitude airspace, which covers airspace representation, planning, monitoring, and scheduling, as detailed below:

- **Spatiotemporal Resources:** Since low-altitude flight tasks operate under varying temporal and spatial conditions,

finding the optimal flight paths and time windows within limited resources is crucial [65]. An spatiotemporal management system can dynamically adjust the requirements of flight tasks and enhance airspace utilization efficiency.

- **City Information Modelling (CIM):** By integrating geographic information, traffic networks, and building data, a digital model is created to help aircraft understand urban environments [66]. This enables intelligent path planning and obstacle avoidance, and provides real-time environmental data to enhance flight precision and safety.
- **Knowledge/Rule Base:** The knowledge base [67] compiles information on low-altitude flight, including operational guidelines, airspace regulations, and meteorological data, to provide real-time decision support. The rule base [68] defines standards and regulations to ensure compliant operations and prevent conflicts.
- **Airspace/Flight Management:** Airspace management plans and monitors low-altitude airspace to prevent collisions or conflicts between aircraft [69]. Flight management handles path planning, task execution, and flight status monitoring for aircraft, which ensures timely task completion and compliance with safety regulations [70].

3) **Multi-Collaboration and Service Assurance Layer:** The third layer of the LAE network architecture meets the regulatory needs of functional departments and the operational requirements of enterprises. By fully integrating digital management and services, this layer empowers governments, air traffic control departments, operators, and businesses to ensure efficient, safe, and compliant low-altitude flights. Moreover,

this layer promotes collaboration through information sharing, real-time monitoring, and intelligent analysis to ensure smooth task execution within the LAE networks.

Low-Altitude Supervision System: The system links military and civil aviation regulatory agencies with government management platforms, which focus on airspace policy formulation, regulatory enforcement, and compliance oversight. Specifically, the core roles of such systems include real-time monitoring of airspace activity, evaluating the compliance of flight operations, and ensuring airspace safety and order via effective management. Based on this, the system mainly consists of two key modules:

- **Identity Authentication:** Digital certificates, biometrics, and aircraft codes are used to verify the identity of the aircraft and its crew. Moreover, the system cross-references databases to confirm qualifications and ensure compliance with airworthiness standards [71]. Only the certified aircraft and personnel are granted takeoff clearance, thus preventing illegal flights from the outset.
- **Violation Handling:** Illegal flights are met with swift legal action, including thorough investigation, evidence collection, and penalties ranging from fines to license suspension or even aircraft confiscation [76]. Moreover, violations are made public to deter others and uphold the integrity of low-altitude airspace.

Low-Altitude Flight Service System: The system is responsible for the overall organization and coordination of low-altitude aircraft tasks at the planning and management stage. Specifically, system-level strategies and adaptive planning mechanisms are applied to ensure efficiency, stability, and responsiveness in supporting subsequent task execution, thereby establishing a solid foundation for successful task completion. The system consists of two main modules:

- **Task Scheduling:** This process focuses on the dynamic assignment of tasks to aircraft and the determination of their execution sequences based on real-time factors such as task urgency, spatial distribution, and platform constraints (e.g., battery level or payload capacity). These strategies aim to enhance responsiveness, balance operational load, and prevent system bottlenecks [77], [78].
- **Resource Coordination:** This function manages the deployment and redistribution of aircraft, energy supplies, and support infrastructure. Through intelligent forecasting [79] and cross-departmental resource sharing [80], the module ensures timely availability of essential resources. Such coordination reduces redundant allocations and improves resilience under fluctuating task demands.

Low-Altitude Flight Control System: The system exercises direct control over aircraft during task execution. In dynamic low-altitude environments, flight control is achieved through continuous state monitoring, proactive conflict and obstacle detection, and the issuance of in-flight adjustment commands to maintain safe navigation and ensure task reliability. The system is structured around two essential components:

- **Trajectory Optimization:** This generates efficient flight trajectories by adjusting parameters such as altitude, heading, and waypoint timing in response to operational

constraints and environmental conditions [81]. The goal is to minimize flight duration, reduce energy consumption, or maximize task coverage, thus supporting optimal in-air task performance [82].

- **Obstacle Avoidance:** As a core safety function, this enables aircraft to detect obstacles such as buildings, terrain, or nearby aircraft through onboard sensing [83]. Subsequently, the aircraft perform timely and localized path adjustments in response to these threats, thereby maintaining safe separation while ensuring efficient and stable task execution.

C. Key Findings and Insights

Based on the above content, it can be concluded that LAE networks are built upon a multi-layered architecture supported by advanced standards, intelligent systems, and robust infrastructure to enable scalable, autonomous, and reliable UAV operations in low-altitude airspace. In the following, we outline key issues that require further research.

- **Modular Design:** A core trend in LAE networks is the rise of modular platforms that enable low-altitude aircraft to switch task functions rapidly without full system reconfiguration. Compared to traditional tightly coupled architectures, modern platforms decouple task payloads such as sensors, communication units, or control modules into pluggable components [84]. This enables in-flight reconfiguration based on real-time needs, which supports efficient adaptation to scenarios such as patrol, logistics, or emergency response. As this paradigm grows, standardized interfaces and blockchain-based management help ensure secure integration, regulatory compliance, and component traceability [85].
- **Edge Intelligence:** Another key insight lies in the transition from centralized cloud computing to onboard AI deployment in low-altitude aircraft. In bandwidth-limited or interference-prone environments, real-time decisions must be made locally. To address this, LAE systems embed lightweight AI models for core functions such as local sensing, flight behavior adjustment, and collision avoidance, greatly reducing latency and external dependence [86]. These models use neural architecture search and quantization to fit edge devices, while federated learning allows aircraft to improve local models collaboratively without sharing raw data [87].
- **Infrastructure Awareness:** Traditional routing algorithms for low-altitude aircraft often focus on minimizing distance or energy use while overlooking the role of critical ground infrastructure, such as charging stations, maintenance hubs, and emergency landing zones. In dense operational environments, routing must adapt to airspace dynamics and infrastructure constraints. Through DT integration of ground facilities, routing plans can dynamically account for real-time loads and queuing delays [88]. This enables a shift from isolated aerial optimization to system-level coordination that improves resilience and resource efficiency.

- *Context Adaptation*: Effective task execution in LAE networks requires aircraft to understand and adapt to context such as weather shifts, population density, and regulatory changes. To adapt accordingly, instead of reacting passively, aircraft equipped with semantic modeling and rule-based reasoning can revise task priorities and update planned trajectories in response to current constraints [89]. By combining geographic data, time factors, and airspace policies into lightweight inference engines, aircraft gain the ability to adapt strategies in real time. Therefore, this context-aware capability supports safer and more intelligent responses to dynamic airspace.
- *Explainable Autonomy*: As LAE networks expand into urban and policy-sensitive areas, system transparency is essential for earning trust from operators, regulators, and the public. Aircraft must perform safely and provide understandable justifications for their actions. Explainable autonomy frameworks enable human observers to interpret the logic behind aircraft behavior through interfaces that visualize decision factors [90]. These explanations enhance accountability and strengthen the credibility of automated operations, particularly in environments that demand social oversight and regulatory compliance.

III. ENABLING TECHNOLOGIES FOR LAE NETWORK DEVELOPMENT

The development of LAE networks is based on three core technologies: (1) integration of communication, sensing, and intelligent computing, (2) collaboration of positioning, navigation, and surveillance, and (3) fusion of flight control with airspace management. This section provides a detailed overview of these technologies.

A. Integration of Communication, Sensing, and Intelligent Computing

This integration provides essential technical capabilities for low-altitude aircraft. Communication facilitates reliable real-time data exchange, sensing accurately detects the surrounding environment, and intelligent computing leverages big data and GAI for rapid analysis and decision making. The synergy of these technologies enables aircraft to quickly adapt to environmental changes, optimize flight paths, and efficiently execute tasks.

1) *Next-Generation Communication*: To support large-scale and dynamic low-altitude environments, LAE networks require communication systems with enhanced reliability, responsiveness, and adaptability.

• **5G-Advanced (5G-A)**: As communication demands grow, the conventional 5G networks face challenges such as spectrum constraints, limited coverage, and difficulties in latency control, particularly for the bandwidth, high-reliability, and low-latency applications. To address more complex future needs, 5G-A [91] has emerged as an enhancement and expansion of 5G, which not only inherits key features of 5G, such as eMBB, URLLC, and massive machine type communication (mMTC), but also further integrates new technologies such as real-time broadband communication (RTBC), uplink

centric broadband communication (uCBC), and harmonized communication and sensing (HCS) [92]. These technologies can effectively support high-demand applications, including massive IoT [93], autonomous driving [94], and industrial automation [95].

In the 5G-A technological architecture, millimeter-wave (mmWave) communication plays a crucial foundational role. As traditional spectrum resources continue to dwindle, the mmWave frequency band (i.e., 30 GHz to 300 GHz) has become a vital technology, which offers large bandwidth and high data rates, thus making it essential for the applications such as ultra-high-definition video streaming and virtual reality [96]. However, mmWave also faces several challenges, including high propagation loss and poor penetration, which require precise beamforming techniques or dense network layouts [97] to compensate for these limitations and ensure stable signal transmission. In this context, massive multiple-input multiple-output (MIMO) [98] has been widely applied in 5G-A. By utilizing a large number of antenna elements, massive MIMO enables spatial multiplexing and beamforming to enhance signal quality. When combined with mmWave communication, massive MIMO not only mitigates the limitations of mmWave, but also fully leverages its abundant spectrum resources, thereby significantly boosting network capacity and data transmission rates [99].

In addition to the traditional techniques, AI and machine learning (ML) approaches have shown great potential in enhancing the performance of mmWave massive MIMO systems. These technologies not only help optimize the beamforming and channel estimation, but also enable real-time and automated network control in dynamic environments. For instance, a deep learning-based hybrid beamforming and channel estimation approach was proposed in [100], which focuses on improving spectral efficiency and lowering computational costs, but does not consider hardware constraints such as low-resolution phase shifters (PSs) and analog-to-digital converters (ADCs). To this end, [101] introduced a neural hybrid beamforming and combining (NHB) scheme based on autoencoders, thereby enhancing bit error rate (BER) performance by up to 4 dB and improving flexibility under low-resolution hardware. Moreover, [102] proposed an efficient aided graph neural network (GNN) with hierarchical residual learning for beamspace channel estimation, which demonstrates strong scalability and robustness by maintaining low normalized mean squared error (NMSE) under -5 dB signal-to-noise ratio (SNR) and large antenna array configurations.

With the evolution of access technologies, 5G-A also introduces non-orthogonal multiple access (NOMA), which enables more efficient user access through flexible power allocation and multi-user multiplexing strategies on shared frequency resources, further boosting network capacity and stability. For example, the authors conducted a detailed capacity analysis of NOMA in mmWave massive MIMO systems [103], which shows a 240% capacity gain over long-term evolution (LTE). However, their analysis relies on simplified single-path assumptions and omits multi-cell interference. To improve practical performance, the authors proposed a practical optimization framework for uplink mmWave MIMO-NOMA,

integrating clustering, beamforming, and power allocation to achieve a 54.6% spectral efficiency gain over orthogonal multiple access (OMA) [104]. Nevertheless, the method assumed perfect channel state information (CSI) and focuses solely on uplink performance. Moreover, NOMA was applied to beamspace MIMO systems, where lightweight user clustering and lens antenna arrays were combined with a clustering-based NOMA throughput optimization algorithm [105]. Compared to traditional beamspace-MIMO-NOMA, this approach achieves 37% and 26% improvements in spectral and energy efficiency, respectively, while maintaining low computational complexity.

· **GAI-based Beamforming:** Beamforming plays a central role in LAE networks by enhancing air-to-ground link quality, enabling secure cooperative transmission via virtual antenna arrays, and facilitating multi-task inference in edge-intelligent environments. To further improve the adaptability and robustness of beamforming in complex scenarios, researchers have proposed GAI-based methods with notable progress. For example, the authors in [106] developed a generative diffusion model (GDM)-enabled twin delayed deep deterministic policy gradient (TD3) approach to dynamically generate UAV excitation currents and position parameters, so that an average secrecy rate of 7.24 bps/Hz is achieved, and flight energy consumption is reduced to 1879.85 J per step with 8 UAVs. Similarly, the authors used the GDM-TD3 method to optimize collaborative UAV beamforming, thus achieving the highest worst-case secrecy rate and lowest per-step energy consumption [107]. However, the complexity of these models may lead to higher training costs. For multi-task intelligent inference, the authors designed an online policy generator based on diffusion models to jointly optimize batch scheduling and beamforming design [108], thereby increasing the inference throughput by up to 40.25% compared to existing methods under asynchronous task arrivals. In addition, [15] explored a robustness-oriented beamforming algorithm based on a mixture of expert (MoE)-transformer architecture, which achieves more than a 44% increase in the worst-case secrecy rate under adverse channel conditions.

· **GAI-enabled Semantic Communication:** Despite the physical-layer advances, further progress in communication efficiency and coordination within LAE networks requires innovations in information representation and transmission. Semantic communication addresses this by conveying task-relevant meaning instead of raw data. Building upon this, GAI-enabled semantic communication enhances transmission effectiveness and ensures higher content fidelity across diverse wireless scenarios. For instance, in [109], a generative adversarial network (GAN)-based system was proposed for intelligent transportation, which enables a 93.45% reduction in communication volume while maintaining high-quality image reconstruction. Moreover, the authors in [110] developed a system combining swin transformer encoding and diffusion model decoding, thereby achieving peak signal-to-noise ratio (PSNR) improvements of 17.75% in additive white Gaussian noise (AWGN) channels and 20.86% in Rayleigh fading channels. Meanwhile, this method addresses the perceptual degradation often observed under low SNR conditions, which cannot be fully captured by traditional distortion-based metrics. For

task-driven applications, [111] designed an adaptive semantic communication framework that transmits only task-oriented features, so that performance in object detection and instance segmentation is improved. In addition, [112] focused on wireless video transmission using a stable diffusion-based architecture, which results in up to 69% improvement in PSNR while reducing mean squared error (MSE) and Fréchet video distance (FVD) by up to 52% and 38%, respectively. However, reliance on frame interpolation and generative diffusion may limit temporal consistency and real-time responsiveness in highly dynamic scenarios.

· **Low Earth Orbit (LEO) Satellites:** Due to the relatively low orbital altitude (i.e., ranging from 500 to 2000 kilometers), LEO satellites experience lower latency and path loss, leading to more stable communication signals, which is especially suitable for low-altitude aircraft operating across wide areas. To fully leverage the potential of LEO satellites, several innovative technical solutions have been proposed. For instance, the LEO satellite selection-based computation offloading algorithm achieved an average total delay of 42.76 ms and improved resource utilization to 66.67% through optimizing satellite selection [113]. In terms of channel allocation, an improved artificial bee colony (IABC) algorithm was introduced to mitigate beam interference in LEO satellite networks, optimize channel utilization, and reduce propagation delay [114], thereby enhancing the system throughput to 112.86 Mbps. Furthermore, the adaptive coverage enhancement approach improved the success rate of random access and reduced power consumption [115], significantly boosting the performance of Narrowband IoT in LEO satellite networks. This approach also reduces the performance gap by approximately 58% compared to the traditional 3GPP model. Despite these advances, as the number of aircraft and the task complexity grow, frequent handover of LEO satellites may result in transient communication interruptions, increased latency in data exchange, or delays and losses in control command delivery.

To address these challenges, several representative mitigation strategies have emerged in recent years. For instance, a deep Q-network (DQN)-based method was designed, in which aircraft were treated as agents to optimize handover and satellite selection [116], thereby achieving up to 4.55% higher throughput and 11.94% lower latency via reduced hop distance. Moreover, the authors introduced a GNN-based distributed handover strategy, which allows user equipment to independently select target satellites and employs ACK-based admission control to enforce load balancing [117], so that handover failures in congested user scenarios can be mitigated. Another promising solution was proposed in [118], which integrates multi-agent reinforcement learning (MARL) and game theory to reduce handover frequency by over 16%, decrease blocking rates by 18%, and improve system utility by up to 48%. In addition, the authors developed a deep reinforcement learning (DRL)-based handover optimization protocol (DHO), which simplifies the conventional handover process by skipping the measurement report phase and leveraging locally observable satellite dynamics [119]. Simulation results show that DHO reduces access delay by up to 6.86 times,

while improving handover success rate and lowering collision probability in dense LEO networks.

2) *Integrated-Active and Passive Sensing*: Sensing technologies gradually evolve toward the integration of collaborative active sensing and non-collaborative passive sensing [120].

• *Collaborative Active Sensing*: On the one hand, collaborative active sensing relies on the cooperation of multi-modal sensing technologies, which integrate various sensors, such as radar, LiDAR, vision, and infrared to comprehensively perceive the environment from multiple dimensions. The complementarity of these sensors overcomes the limitations of individual sensors [121], [122], which significantly improves the sensing accuracy and enhances target detection, recognition, and tracking precision. For example, the authors designed a feature registration and complementary sensing network (FRCPNet) [123], which aligns multi-level features at the pixel level and enhances the semantic correlation between modalities, thereby improving the accuracy of salient object detection in combined RGB and thermal images. Moreover, in [124], a visual transformer-based model (ViTAL-TAPE) that combines photometric and radiometric data was introduced to overcome the limitations under varying lighting and viewing conditions, ultimately achieving high-precision landing detection with an accuracy of 0.01 meters.

On the other hand, collaborative active sensing benefits from integrating sensing and communication to enhance both the accuracy and timeliness of environmental perception. In this scheme, active sensing collects real-time data, which is efficiently shared via communication to improve system responsiveness. For instance, in [125], the authors proposed a joint sensing-communication UAV network that synchronizes radar sensing and data communication through beam sharing to strengthen the collaborative sensing capacity. In another study [126], the authors combined secondary surveillance radar with transponder-equipped air obstacles to enable high-risk target identification with 0.997 accuracy within a 10 km range by sharing altitude and identification code information. Furthermore, ISAC is emerging as a key approach, aiming to simultaneously achieve high-efficiency communication and high-precision sensing within shared spectrum resources. To support this vision, GAI offers promising solutions with its powerful capabilities in data modeling, inference, and generation. Specifically, the authors in [127] proposed a diffusion model-based direction-of-arrival estimation method for ISAC, which achieves a mean square error of just 1.03 degrees under near-field conditions. Moreover, [128] applied conditional GANs to channel estimation in reconfigurable intelligent surface (RIS)-assisted ISAC systems, so that modeling precision and training efficiency were improved. However, the generalization capability under dynamic RIS configurations or non-stationary channels still requires further investigation. In addition, a secure sensing framework for ISAC was introduced in [129], which combines discrete and continuous diffusion models to generate safeguarding signals for authorized devices, thereby reducing the sensing accuracy of unauthorized devices by approximately 70%. Nevertheless, the computational overhead introduced by dual diffusion models

may hinder their real-time deployment in resource-constrained edge environments.

• *Non-Collaborative Passive Sensing*: Non-collaborative passive sensing involves utilizing existing wireless communication signals (e.g., Wi-Fi or 4G/5G signals) to detect the surrounding environment. Unlike active radar, passive sensing captures reflected waves from signals emitted by wireless communication systems to gather environmental information, without the need for dedicated transmission signals [130]. The main advantage of this technology is that it does not interfere with existing communication systems and requires no additional hardware, which makes it ideal for privacy-preserving and cost-effective monitoring applications [131]. However, weak reflected signals in passive sensing are susceptible to multi-path effects, static clutter, and noise. Additionally, extracting useful information from these complex signals and accurately detecting moving targets without active control remains a challenge.

Consequently, the integration of active and passive sensing will become the core of intelligent sensing systems. By complementing each other, the system can leverage active radar for high-precision distance and velocity data, while also using the environmental changes detected by passive sensing to form a richer and more accurate perception. For example, in LAE applications, active radar measures real-time distance between UAVs and other aircraft, thus providing precise flight trajectory information, while passive sensing aids in detecting and tracking surrounding ground targets, such as vehicles or buildings. This collaboration not only improves the perception accuracy and robustness, but also optimizes the resource utilization. However, fully exploiting the benefits of this collaboration requires overcoming challenges such as signal interference, real-time system coordination, and multi-modal data fusion.

3) *GAI-driven Computing*. Traditional AI plays a crucial role in mobile edge computing (MEC) and cloud-edge-end collaboration [132]. However, it faces several challenges when applied to large-scale scenarios. These include difficulty handling unpredictable environments, increased latency when processing large data volumes, and loss of key information during high-dimensional feature extraction. To address these issues, GAI has emerged as a promising solution to improve decision-making in LAE networks.

• *GAI-driven MEC*: In MEC systems, GAI [133] leverages its powerful modeling and contextual reasoning capabilities to optimize task offloading, resource allocation, and other operations in dynamic environments, as shown in Fig. 3 Part A, thereby enhancing decision quality and system adaptability. Among various GAI paradigms, GDMs have shown particular promise in this regard, as they support reinforcement learning (RL) through synthetic data generation, policy refinement, and environment simulation. For example, the authors employed GDMs to synthesize velocity states from neighboring UAVs [134], thereby alleviating data scarcity from communication delays while improving training efficiency in trajectory planning and obstacle avoidance. Moreover, a conditional diffusion framework with binary masks was proposed to enable the automatic generation of annotated UAV images [135], which enhances YOLOv7 detection

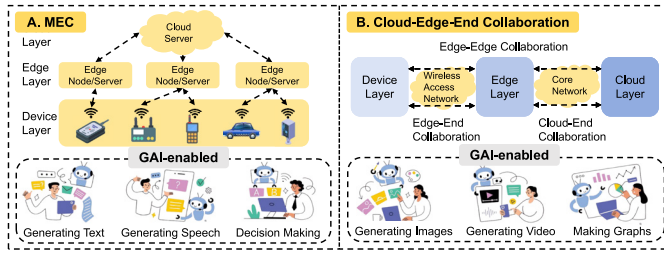


Fig. 3. GAI-driven computing in LAE networks. *Part A* presents an MEC-based architecture, where GAI supports applications including text generation, AI chatbots, and decision-making. *Part B* illustrates a cloud-edge-end collaborative framework, where GAI facilitates functions such as image generation, video generation, and graph creation.

performance by 42% in AP@0.5:0.95 and supports scalable dataset expansion. In policy learning, the deep diffusion soft actor-critic (D2SAC) framework was introduced [136], which reaches baseline performance in 190 training steps, with utility of 494, crash rate of 1.1%, and training time of 1.3 hours. Similarly, GDMs were integrated into a TD3 framework for UAV-assisted vehicle twin (VT) migration, thus the workloads of roadside units (RSUs) by up to 52% and energy costs of UAVs by 15% [137]. For environment simulation, the adaptive diffusion environment for policy transfer (ADEPT) framework was proposed [138], which achieves a 0.87 navigation success rate in simulation and completes 1.3 km field trials without fine-tuning, thereby improving zero-shot transfer and policy robustness. In addition, the authors introduced the adversarial environment design via regret-guided diffusion models (ADD) [139], so that an 85% success rate in partially observable navigation is achieved and 72% performance is maintained with half the environment steps. However, limitations remain in the convergence guarantees, approximate regret estimation, and delayed feedback between policy and environment.

Complementary to GDMs, large language models (LLMs) have emerged as another key paradigm of GAI in LAE networks. Specifically, prompt engineering offers a lightweight solution that requires no modification of the model parameters. This approach is based on natural language instructions and demonstrations, so that the model can fully activate its existing knowledge and reasoning capabilities to accomplish complex tasks. For instance, the in-context learning (ICL) approach [140] allows a model to utilize historical network configurations or optimization strategies as contextual input. Moreover, the chain-of-thought prompting technique [140] enhances intermediate reasoning transparency, which supports logically demanding tasks without increasing inference cost. Building upon these features, ICL has been effectively applied in emergency scenarios. In the ICL-based data collection scheduling framework, the UAV collects the real-time status of the ground sensors, such as queue lengths and channel conditions. This information is then converted into structured natural language task descriptions and delivered to the LLM, which generates updated scheduling strategies based on the current context. Once the strategy is executed, performance feedback, such as packet loss, is incorporated into the task description. This updated context serves as the basis for the

next inference cycle, allowing the system to refine its decisions over time and respond effectively to evolving environmental conditions. As such, this closed-loop mechanism enhances operational robustness and makes the ICL particularly suitable for low-altitude UAV systems operating under constrained communication and limited computational resources.

Additionally, prompt engineering [140] exhibits a structured ability to handle complex tasks, which enables effective support for multi-stage reasoning and autonomous decision-making. This capability aligns well with the characteristics of LAE networks, where task dependencies often involve multiple variables and layered processes. Specifically, the chain-of-thought prompting technique [140] guides the model to explicitly present intermediate reasoning steps in a progressive manner, so that the logical clarity and problem-solving ability are significantly enhanced. In addition, the prompt-based planning strategy [140] allows large-scale network operations to be broken down into a series of manageable subtasks, which fits scenarios that require sequential execution such as network planning, deployment, and dynamic reconfiguration. Beyond this, the self-refinement mechanism [140] permits the model to generate feedback from its own outputs and revise prompts iteratively without human intervention. Consequently, this integrated prompting framework provides an effective solution for LAE wireless systems that require real-time adaptation and continuous learning, while highlighting the unique strengths of prompt engineering in enabling system intelligence and long-term sustainability.

GAI-driven Cloud-Edge-End Collaboration: To further enhance the adaptability and responsiveness of the system, the cloud-edge-end collaboration architecture has emerged to provide a more flexible solution, as shown in Fig. 3 *Part B*. In this architecture, GAI can be integrated with extended reality (XR) systems to enhance content generation, system adaptability, and intelligent interaction. To be specific, LLMs enable natural language communication in XR environments so that non-expert operators can control UAV tasks through semantic commands, which significantly lowers the operational threshold. Furthermore, GAI dynamically generates task scenarios, training protocols, and simulated environments, thereby enriching virtual reality (VR)-based instruction and mixed reality (MR)-based rehearsals with greater diversity and realism. In addition, retrieval-augmented generation (RAG) mechanisms allow LLMs to incorporate relevant external knowledge [141], which improves contextual understanding in augmented reality (AR) interfaces for navigation and emergency response [142]. Beyond this, GAI-powered intelligent agents perform subtasks such as target identification, route optimization, and situational forecasting within XR systems so that task automation and system autonomy can be substantially enhanced. Additionally, GDMs and similar visual generation techniques allow XR platforms to construct detailed 3D environments, reconstruct post-disaster areas, and synthesize realistic visual data, which supports the development of more robust perception models [142]. Despite these advancements, GAI-augmented XR systems still face critical challenges, including privacy risks from sensitive data exposure, stringent

security requirements for content transmission, ethical concerns over opaque content generation, and heavy system loads due to high data intensity.

In response to some of these challenges, the integration of GDMs within DT systems is emerging as a promising approach, which demonstrates significant advantages in data augmentation, behavior modeling, and intelligent decision-making. For example, in UAV communication, GDM was served as a generative transmitter and receiver to handle compression, denoising, and reconstruction [143], thereby improving the reliability and efficiency of data exchange over noisy channels. To address data scarcity, GDM was applied to generate synthetic datasets that accurately reflected real-world dynamics [144], which expands the training resources and enhances the generalization capability of RL models. However, ensuring adequate coverage of edge-case behaviors in the generated data remains a significant challenge. Moreover, in task coordination, the authors proposed a return-conditioned GDM to guide solution generation based on target outcomes [134], so that training efficiency and inference accuracy are improved in tasks such as path planning and resource allocation. Nevertheless, heavy reliance on well-designed reward functions may hinder the generalization of this method across diverse tasks.

In addition to these developments, it is important to recognize that the deployment of GAI in practical LAE networks still faces several limitations that deserve careful consideration. Specifically, large-scale generative models typically require extensive training data and computational resources, which may be difficult to support on resource-constrained edge nodes [145]. Moreover, the poor interpretability of GAI-generated outputs may hinder reliability and traceability in safety-critical applications [146]. Furthermore, the highly dynamic nature of edge environments can lead to poor generalization when deployment contexts diverge from training scenarios [145]. In addition, reliance on diverse and distributed data sources increases the complexity of ensuring data privacy and security, particularly in critical or sensitive tasks [147].

In conclusion, Table V illustrates the applicability of communication, sensing, and intelligent computing in LAE networks. By adopting these advanced technologies, efficient data transmission and stable connectivity, comprehensive environmental sensing, and intelligently assisted decision-making can be achieved. However, relying solely on individual technologies or limited combinations of them is no longer sufficient to meet the growing demand.

Numerous studies have explored the integration of these three technologies. For example, [149] proposed a unified ISCC framework, aiming to optimize the limited system resources in 6G wireless networks. In [150], the authors investigated a UAV-enabled ISCC (U-ISCC) framework and analyzed the trade-off between the computational capability and beam-sensing gain. Moreover, a novel wireless scheduling architecture was introduced [148] (Fig. 4) that jointly optimizes the coordinated gains of sensing, communication, and computing to address heterogeneous demands in ISCC. In the context of IoT and cellular networks, [151] examined the ISCC challenges in beyond-5G cellular IoT through two joint beamforming design algorithms. Furthermore, in [152],

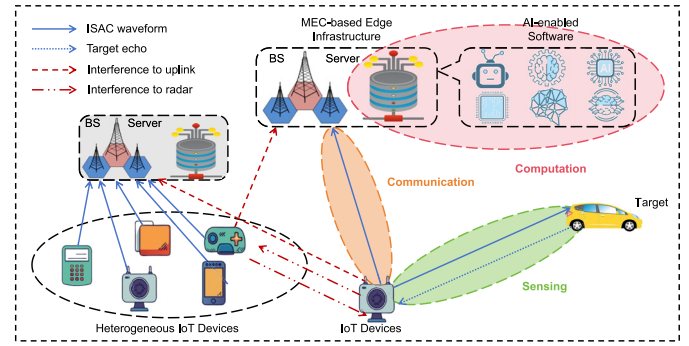


Fig. 4. Proposed multi-user wireless network architecture under the ISCC paradigm in [148]. Specifically, the BS utilizes ISAC waveforms to integrate communication and sensing while collaborating with IoT devices for computational tasks. Moreover, MEC-based edge infrastructure and AI-enabled software provide intelligent decision support.

the authors presented AirComp as a communication approach and designed two MIMO ISCC over-the-air (ISCCO) schemes (i.e., separated and shared) to enhance performance. In edge computing, [153] studied a multi-device edge AI system that integrates AI model splitting with ISAC to deliver low-latency intelligent services. In addition, [154] developed a NOMA-assisted joint communication, sensing, and multi-tier computing framework. Despite these developments, several challenges persist in real-world LAE applications. Specifically, the rapid growth of the IoT devices and the limited communication resources complicate the access and transmission of large-scale devices. Moreover, the air interface between UAVs and ground control systems in LAE networks is inherently dynamic and open, which increases the complexity of achieving resilient coverage and secure communication in a 3D space. As a result, the seamless integration of aerial vertical coverage with terrestrial cellular networks has become a pressing issue.

To address this, the careful planning of spectrum resources and the introduction of dynamic interference management mechanisms are essential to improve spectrum efficiency, particularly in the context of co-existence with existing 5G infrastructure. For example, the study [155] demonstrated that RIS-assisted joint beamforming methods can intelligently control spatial degrees of freedom under limited power constraints, which effectively suppresses co-channel interference in spectrum reuse scenarios and reduces sensing errors by more than 46.7%. Moreover, the simultaneously transmitting and reflecting surface (STARS)-assisted orthogonal frequency division multiple access (OFDMA) ISCC network was introduced to jointly optimize subcarrier allocation and reflection/transmission beamforming [156], so that sensing, communication, and computing functions share spectrum efficiently in wide-bandwidth scenarios. Furthermore, the authors in [157] proposed a two-stage GAI framework, which uses a reconstructive masked autoencoder (RecMAE) to recover spectrum maps from sparse and time-varying data, followed by a multi-agent diffusion policy (MADP) to optimize UAV trajectories. Compared to the interpolation and deep learning baselines, the approach reduces the reconstruction error by

TABLE V
SUMMARY OF THE INTEGRATION OF COMMUNICATION, SENSING, AND INTELLIGENT COMPUTING TECHNOLOGY

	Technologies	References	Pros & Cons
Next-Generation Communication	5G-A	[100]–[105]	✓ Increasing network capacity and data transmission rates ✓ Achieving automated network optimization in dynamic environments ✗ Requiring extensive resources for mmWave beamforming and network layout
	GAI-based Beamforming	[15], [106]–[108]	✓ Improving secrecy and energy efficiency via generative policy optimization ✓ Enhancing multi-task inference throughput in dynamic edge environments ✗ Increasing training complexity due to large-scale generative models
	GAI-enabled Semantic Communication	[109]–[112]	✓ Reducing transmission volume while preserving semantic fidelity ✓ Improving task-specific performance through adaptive feature extraction ✗ Limiting temporal consistency and real-time response in dynamic scenarios
	LEO satellite	[113]–[119]	✓ Providing stable communication signals for wide-area coverage ✓ Improving efficiency, optimizing resources, and boosting throughput ✗ Frequent switching of LEO satellites potentially complicating system management
Integrated-Active and Passive Sensing	Collaborative Active Sensing	[123]–[129]	✓ Aligning sensing and communication for accurate and responsive perception ✓ Applying GAI for high-precision estimation and secure signal generation ✗ Introducing computational overhead that limits real-time deployment on edge devices
	Non-Collaborative Passive Sensing	[130], [131]	✓ Avoiding interference with communication systems, and reducing hardware needs ✓ Enabling privacy-preserving and cost-effective monitoring applications ✗ Struggling with weak reflected signals, multi-path effects, and noise
GAI-driven Computing	GAI-driven MEC	[134]–[140]	✓ Enabling efficient edge learning and decision-making via diffusion and prompting ✓ Improving policy robustness and zero-shot adaptability without retraining ✗ Lacking convergence guarantees and adding overhead in regret-driven models
	GAI-driven Cloud-Edge-End Collaboration	[134], [142]–[144]	✓ Facilitating intuitive interaction and task automation via GAI-XR integration ✓ Enhancing data efficiency and decision-making through GDM-enabled DT systems ✗ Exposing privacy risks and imposing high data and security burdens

57.35% and 88.68%, thereby improving the accuracy of spectrum sensing and efficiency of deployment under interference.

In addition to spectrum planning and interference control, it is equally important to enhance system intelligence and adaptability to dynamic environments, so as to effectively cope with the inherent dynamism and instability in LAE network deployments. For instance, [158] proposed a GAN-enhanced TD3 algorithm for double-active RIS-assisted ISAC networks, which jointly optimizes transmit, reflection, and receive beamforming to improve both sensing accuracy and communication robustness. Moreover, the authors designed a GDM-based deep deterministic policy gradient (DDPG) algorithm to solve a multi-objective optimization problem, which integrates communication, sensing, and energy efficiency [159], so that UAV trajectory planning and beamforming configuration can be efficiently optimized. Furthermore, in [160], the authors presented a DT-assisted offline DRL framework that incorporates a conditional GAN to simulate environmental transitions, enabling more efficient beam selection and reducing online training overhead by 80%. Inspired by these advances, future LAE network deployment can leverage ISCC capabilities to construct a dynamic radio environment map that captures spectrum states and interference patterns, while GAI enables spectrum awareness, adaptive modeling, and predictive optimization in response to dynamic and uncertain conditions. Together, these technologies enable the joint optimization of multiple access and interference management, ultimately facilitating intelligent spectrum sharing and resource scheduling in air-ground integrated environments.

B. Collaboration of Positioning, Navigation, and Surveillance

This collaboration forms the foundation for flight stability and precise operations in the LAE networks. Positioning

ensures accurate location tracking in complex environments, navigation provides optimal route planning, and surveillance continuously monitors the airspace, thus identifying potential risks in real time. The coordination of these technologies greatly increases the situational awareness of the aircraft and reinforces the safety defenses of the system.

1) *High-Accuracy Positioning*: The GNSS is the cornerstone of modern positioning technology. Specifically, GNSS determines accurate 3D positioning (i.e., longitude, latitude, and altitude) by receiving radio signals from multiple satellites and calculating the distance between the receiver and each satellite, along with precise time synchronization. Common GNSS systems include GPS, GLONASS, Galileo, and BeiDou. They have played a crucial role in a wide range of applications, such as intelligent transportation systems (ITS) with connected autonomous vehicles (CAVs) [161], [162], [163], as well as in complex environments such as urban canyons [164], [165], [166].

• *Improved GNSS*: To further enhance the performance of GNSS, researchers have explored innovative technologies in various fields, such that significantly increasing the accuracy and reliability of the GNSS positioning. In the agricultural vehicle sector, [167] proposed a collaborative positioning algorithm that integrates vehicle motion state data and GNSS observations by using an extended Kalman filter (EKF), thus boosting the performance of standalone GNSS positioning. Subsequently, based on Bayesian theory, the author combined GNSS data, ultra-wideband (UWB) ranging data, and inter-vehicle relative positioning information to further enhance the positioning accuracy. Moreover, the authors in [168] developed a generalized theoretical framework for analyzing and measuring vehicle distances in workshops, which enables precise distance measurement by exchanging GNSS observation values. For complex urban environments, challenges such as

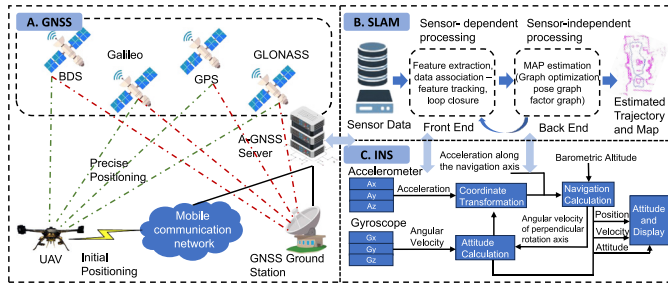


Fig. 5. An integrated UAV navigation and positioning framework. *Part A* presents the GNSS module, which utilizes BDS, Galileo, GPS, and GLONASS for initial positioning, with A-GNSS and ground stations enhancing accuracy. *Part B* outlines the SLAM process, where the front end handles feature extraction and data association, while the back end performs map estimation and pose optimization. *Part C* depicts the INS module, which leverages accelerometer and gyroscope data for attitude and navigation calculations to determine position, velocity, and orientation.

signal blockage, multi-path effects, and NLOS signals may arise. To address these issues, [169] introduced a vision-aided approach that enhances GNSS positioning by identifying and eliminating NLOS satellite signals in urban environments. Furthermore, in [170], the authors presented a joint positioning algorithm that combines Doppler measurements from LEO satellites with GNSS data, so that improving the positioning accuracy to 250 meters. To further support the research, a high-fidelity GNSS urban simulation tool was designed, which uses ray-tracing techniques to replicate signal interference in urban environments [171], thereby helping to precisely analyze signal characteristics and optimize positioning systems.

· **Assisted GNSS (A-GNSS):** The successful application of innovative technologies in the aforementioned fields provides robust technical support for the positioning needs in LAE networks. Additionally, the introduction of A-GNSS technology [172], [173], [174] has become a crucial solution to further boost the positioning performance, as shown in Fig. 5 *Part A*. Specifically, A-GNSS enhances the overall performance of GNSS by utilizing terrestrial mobile communication networks to transmit real-time satellite ephemeris, frequency corrections, and clock calibration data. This allows GNSS receivers to acquire and track satellite signals more quickly, thereby accelerating the signal search process and shortening the time to first fix (TTFF). Furthermore, A-GNSS offers corrections for ionospheric delays, satellite clock errors, and orbital inaccuracies, which minimize the signal propagation errors and significantly improve positioning accuracy. As a result, both the improved GNSS and A-GNSS will play a critical role in meeting the high-precision positioning requirements of low-altitude operations within the LAE networks.

2) **Dynamic-Fusion Navigation:** Integrating single-sensor and multi-sensor systems can enhance positioning accuracy and robustness in GNSS-denied or complex environments.

· **Single-Sensor:** Simultaneous localization and mapping (SLAM) and inertial navigation system (INS) play a crucial role in complex environments, where GNSS signals are unstable or unavailable. SLAM can equip camera [175] or LiDAR [176] sensors to construct environmental maps in real-time and perform self-localization, while integrating GNSS

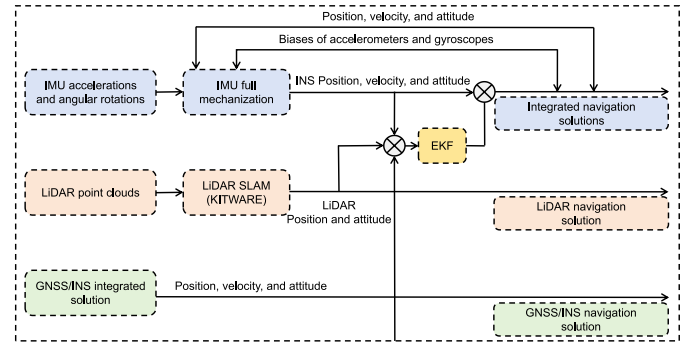


Fig. 6. Overall structure of the INS/LiDAR SLAM LC integration system in [185]. Specifically, the IMU module processes acceleration and angular velocity data to compute the inertial navigation solution. Moreover, the LiDAR SLAM module extracts position and attitude information from LiDAR point clouds, while the GNSS/INS module provides a GNSS-aided inertial navigation solution as an additional reference. Finally, the EKF filter fuses these solutions, updates errors, and generates the final navigation solution. Furthermore, the feedback mechanism corrects IMU mechanization errors through a closed-loop compensation scheme, thereby enhancing the robustness and accuracy of the system.

data to improve positioning accuracy and robustness, as illustrated in Fig. 5 *Part B*. Moreover, INS uses the accelerometer and gyroscope in the inertial measurement unit (IMU) to measure the acceleration and angular velocity of the aircraft in real time, as shown in Fig. 5 *Part C*. This internal measurement approach enables the INS to navigate reliably even in the absence of external signals [177], [178]. However, single-sensor systems often have limitations. Specifically, cameras may struggle in low-light or high-speed conditions, which leads to inaccurate feature detection and difficulties tracking moving objects. Although LiDAR provides precise depth information, it may encounter limitations when dealing with reflective or transparent surfaces, such as windows or water, thereby causing it to produce erroneous measurements or fail to detect objects properly. In addition, INS may experience drift in the absence of external reference points, such as when GNSS signals are weak, unavailable, or obstructed, thus leading to cumulative positioning errors that worsen over time. Therefore, by combining sensor data, the navigation system leverages their strengths to provide more accurate and robust navigation than single-sensor systems.

· **Multi-Sensor:** Several studies have highlighted the benefits of multi-sensor fusion in improving navigation performance. For example, the integration of semantic information, laser point clouds, and GNSS data into the visual SLAM framework (SLG-SLAM) [179] was proposed to address the limitations of existing optimization algorithms in complex driving scenarios. Through experiments on four public datasets (i.e., KITTI [180], multi vehicle event (MVE) [181], M2DGR [182], and complex urban (CU) [183]) and one self-collected dataset [184], SLG-SLAM significantly improves the localization accuracy, with the average absolute trajectory error reduced by 43.50% and 14.91% on the KITTI and MVE datasets, respectively. Moreover, in [185], the author developed an EKF-based INS/LiDAR SLAM navigation system through a robust loose coupling approach, which achieves better localization accuracy than pure INS navigation (Fig. 6).

Furthermore, the authors designed a tightly coupled approach that integrates visual, IMU, and GNSS data to jointly optimize image reprojection, IMU pre-integration, and GNSS measurement errors [186]. Notably, the study highlights that this approach outperforms state-of-the-art visual-inertial SLAM, GNSS single-point positioning, and loose coupling approaches, particularly in environments dominated by low-rise buildings and urban canyon scenarios. Consequently, innovative navigation systems based on multi-sensor fusion are key technologies for achieving high precision, reliability, and autonomy in the future of the LAE networks. However, this fusion may face challenges such as higher computational complexity, sensor noise that affects accuracy, and data synchronization issues, which require further attention in the LAE networks.

3) *Real-Time Surveillance*: Automatic dependent surveillance-broadcast (ADS-B) is a data broadcast-based air traffic surveillance technology, which is designed for real-time monitoring and collision avoidance of low-altitude aircraft [187]. To be specific, ADS-B enables aircraft to periodically broadcast position, speed, heading, altitude, and other data via an onboard transmitter. This information is transmitted through radio frequencies and can be received by other aircraft and ground control stations, thereby allowing real-time tracking of the aircraft. Unlike traditional surveillance methods that rely on radar echoes [188], ADS-B provides precise location data based on the GNSS, with the ability to automatically broadcast, which offers advantages in terms of accuracy and timeliness. In addition, ADS-B features high precision, low latency, and wide coverage, so that it can offer reliable surveillance services over large areas. This makes it particularly suitable for complex low-altitude airspace, so that significantly improving the air traffic management efficiency and enhancing aircraft safety. Based on these advantages, [189] developed a conflict awareness scheme for UAVs that analyzes the ADS-B message flow and format to identify potential conflicts. Next, strategies, such as speed adjustments and direction changes, based on the unscented Kalman filter were applied to resolve conflicts. This approach significantly improves the UAV conflict resolution capabilities, and provides a strong theoretical foundation for the advancement of collision avoidance technologies in UAVs.

• **Encryption and Authentication**: Despite these advantages, the limitations of ADS-B should not be overlooked due to its openness and lack of built-in security mechanisms [190]. Specifically, onboard GNSS receivers are vulnerable to interference or spoofing, and fake ADS-B messages can be injected or tampered with. Moreover, attackers can eavesdrop on broadcast messages or manipulate the ADS-B transponder of aircraft. These security vulnerabilities can lead to the distortion of surveillance data, thereby posing significant threats to air traffic safety. To mitigate the risks associated with ADS-B, several technologies have been proposed in recent years. By effectively authenticating and performing integrity checks on ADS-B messages, unauthorized tampering and message injection can be prevented, thereby protecting the system from attacks. For example, the author in [191] presented an

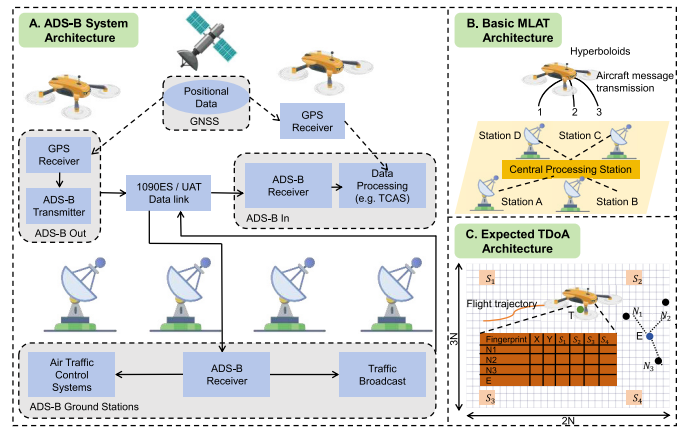


Fig. 7. Air traffic surveillance architectures: ADS-B, MLAT, and expected TDoAs [193]. Part A describes the ADS-B system, which relies on GNSS to provide positional information and facilitates data exchange between aircraft and ground stations through ADS-B Out/In. Part B introduces the MLAT architecture, which enhances surveillance capabilities by using multiple ground-based receivers to measure the TDoA of signals and computes the actual position of the aircraft through a central processing station. Part C presents the expected TDoA architecture, which reduces the reliance on time synchronization by using TDoA fingerprint maps with small grids.

ADS-B message authentication approach based on certificate-free short signatures, which was designed to address security issues in ADS-B systems under low bandwidth and limited data bit conditions. Furthermore, [192] introduced the compatible authenticated bandwidth-efficient broadcast protocol for ADS-B (CABBA). However, this technology requires modifications to the ADS-B protocol, which would result in significant changes for existing aircraft and air traffic management systems.

• **Advanced Time Difference of Arrival (TDoA)**: Additionally, relying solely on encryption and authentication approaches cannot fully address all security threats, particularly in the case of fake data injection. To mitigate this issue, researchers have proposed using multilateration (MLAT) [197], [198] and TDoA [199], [200], [201], [202] techniques to validate the authenticity of position information in ADS-B messages. However, since MLAT suffers from lower accuracy in aircraft positioning, the expected TDoA positioning technique has emerged to address this problem [193]. Specifically, this technique reduces the reliance on time synchronization by using TDoA fingerprint maps with small grids (Fig. 7). However, the positioning accuracy is still less than ideal, and it requires longer computation times. To overcome these limitations, an actual TDoA-based augmentation system (ATBAS) was proposed [194], which uses data from the OpenSky network [193] to train TDoA fingerprint grid models. Experimental results show that the ATBAS framework improves positioning accuracy by 56.93% and 48.86% over MLAT and the expected TDoA technique, respectively.

• **Intelligent Algorithms and Signal Processing**: In response to the threats posed by GNSS spoofing and malicious ADS-B transmitters, researchers also have proposed several innovative protective approaches that combine intelligent algorithms with signal processing techniques to enhance system defense against these attacks. For instance, the author investigated

TABLE VI
SUMMARY OF THE COLLABORATION OF POSITIONING, NAVIGATION, AND SURVEILLANCE TECHNOLOGY

	Technologies	References	Pros & Cons
High-Accuracy Positioning	Improved GNSS	[167]–[171]	✓ Enhancing GNSS positioning accuracy with innovative auxiliary technologies ✓ Supporting precise analysis through simulation tool and generalized framework ✗ Failing to provide accurate positioning in satellite scarcity or signal blockage
	A-GNSS	[172]–[174]	✓ Accelerating signal acquisition and reducing TTFF ✓ Minimizing signal propagation errors and improving positioning accuracy ✗ Relying on the availability and quality of terrestrial communication networks
Dynamic-Fusion Navigation	Single-Sensor	[175]–[178]	✓ Enabling real-time mapping and self-localization in GNSS-denied environments ✓ Providing reliable inertial navigation through internal motion sensing ✗ Suffering from drift or sensor-specific failures under challenging conditions
	Multi-Sensor	[179], [185], [186]	✓ Combining different sensor strengths to overcome single-sensor limitations ✓ Improving localization accuracy and system robustness ✗ Facing higher complexity, sensor noise, and data sync issues
Real-Time Surveillance	Encryption and Authentication	[191], [192]	✓ Reducing the risk of unauthorized message injection and illegal access ✓ Preventing tampering or falsification and improving system security. ✗ Resulting in additional computational overhead and bandwidth consumption
	Advanced TDoA	[193], [194]	✓ Validating the positioning information in ADS-B messages ✓ Reducing dependence on time synchronization ✗ Suffering from lower positioning accuracy, especially in complex environments
	Intelligent Algorithms and Signal Processing	[195], [196]	✓ Strengthening the defense of the system against spoofing attacks ✓ Adapting well to dynamic and complex signal environments ✗ Requiring high data quality and computational resources

a transfer learning-based approach to address the identification of malicious ADS-B transmitters [195]. In addition, the authors applied differential time-series transformation to process ADS-B data and employed non-cyclic autoencoders for attack classification and detection [196]. This approach effectively identifies trajectory modification attacks, thereby improving the ability of the system to guard against GNSS spoofing. However, these techniques require high data quality and computational resources, and in certain complex signal scenarios or cases with significant data discrepancies, the accuracy and generalization ability of the model may be compromised.

Beyond security concerns, the deployment challenges and adaptation issues of ADS-B in low-altitude environments also deserve attention. Specifically, regarding the challenges, the sharp increase in aerial vehicle density, particularly with the proliferation of UAVs and eVTOL aircraft, may lead to broadcast congestion and channel collisions. Unlike well-spaced high-altitude flights, low-altitude traffic tends to be dense and dynamic, which can make it challenging for ADS-B to maintain stable communication due to frequent interference [203]. Moreover, complex urban or mountainous environments often introduce signal occlusion and multipath effects, thereby resulting in intermittent reception or localization errors that may reduce the reliability of GNSS-based positioning. Furthermore, LAE platforms typically experience rapid mobility and frequent topological changes, which may not align well with the static periodic broadcast mechanism of ADS-B [204]. In terms of adaptability, ADS-B was originally designed for crewed aviation, which assumes standardized hardware, ample power supply, and relatively stable flight patterns. In contrast, LAE networks comprise highly heterogeneous and resource-constrained platforms, where limitations in size, weight, and power may restrict the integration of conventional ADS-B modules [205]. Additionally, ADS-B lacks adaptive features such as dynamic broadcast intervals or priority-based message scheduling, which may limit its responsiveness in

decentralized and bursty communication environments. Therefore, effective deployment in LAE scenarios may require not only protocol-level improvements but also reconsideration of key design assumptions to better accommodate lightweight, autonomous, and decentralized operations.

As summarized in Table VI, positioning, navigation, and surveillance each demonstrate significant effectiveness in high-precision positioning, dynamic navigation, and real-time surveillance, respectively. However, most existing research focuses primarily on positioning, navigation, or surveillance, without fully exploring their coordinated integration. To address this gap, recent developments have introduced GAI as a key enabler that integrates sensing, computing, and navigation/surveillance functions. This integration enables intelligent fault detection, safe trajectory modeling, and enhanced operational robustness in UAV systems. For example, the authors in [206] developed the DroneDiffusion framework to capture the multi-modal and stochastic nature of UAV flight dynamics, which was then integrated into control systems to support real-time trajectory adjustments, thus achieving tracking errors as low as 0.019 m in simulation and 0.069 m in real-world tests. Moreover, variational autoencoders (VAEs) were trained to learn normal operation patterns and detect anomalies based on reconstruction error [207], which results in an average detection accuracy of 95.6%. In addition, long short-term memory (LSTM)-based autoencoders were applied to extract spatiotemporal features from sensor data [208], so that the system achieves a 90% detection rate for operational faults and a classification accuracy of up to 99%. As a whole, these efforts reveal the potential of GAI to enhance the coordination and intelligence of future low-altitude aircraft operations.

C. Fusion of Flight Control With Airspace Management

This fusion is crucial for improving the efficiency and safety of low-altitude airspace operations. Flight control optimizes flight paths through intelligent scheduling to ensure

smooth operations in high-density airspace, while airspace management optimizes real-time resource allocation to prevent conflicts and congestion. This integration of these technologies not only improves the operational efficiency, but also maximizes the use of airspace resources, thereby ensuring seamless coordination across multiple tasks.

1) **Advanced-Flight Control:** Proportional integral differential (PID) control [209] is one of the most common traditional flight control approaches. This method maintains aircraft stability by adjusting control inputs based on real-time feedback of attitude and position [210]. However, traditional PID control often struggles to address the challenges posed by nonlinear systems, external disturbances, and parameter tuning when faced with complex and dynamic flight tasks, thus limiting its performance. Moreover, model predictive control (MPC) represents another advanced flight control method that predicts the future state of an aircraft and generates control inputs through optimization to achieve optimal decision-making [211]. Unlike PID control, which focuses on current errors, MPC also considers changes in future states and constraints. For instance, the authors in [212] proposed a general-model dynamic formulation and a two-layered constrained MPC strategy to tackle the trajectory tracking problem for tilt-rotor UAVs. Moreover, a novel nonlinear model predictive control (NMPC) approach was designed to address UAV navigation and obstacle avoidance [213]. While these methods perform well in trajectory tracking and obstacle avoidance tasks, there remains room for improvement in terms of state prediction accuracy and optimization of control decisions. Building on these limitations, AI-based low-level flight control and human-in-the-loop reinforcement learning (HITL-RL)-driven high-level decision-making and coordination strategies have become increasingly essential.

· **AI-based PID:** Adopting AI-based PID control enhances the adaptability and robustness of the system in complex environments. For example, [214] improved traditional PID by integrating a feedforward neural network, which maintains the stability of the quadrotor while addressing hardware limitations and enhancing control accuracy. However, as AI advances, relying solely on PID is inadequate for complex tasks involving external disturbances and internal uncertainties. To address this, a robust intelligent self-adjusting propulsion control approach has emerged. For instance, the authors in [215] developed a hybrid control approach combining PID and intelligent active force control (IAFC) to enhance the disturbance rejection capability and agility performance of a quadrotor UAV. To further enhance robustness, an AI-based iterative learning algorithm (ILAFC) was used to automatically adjust control parameters (Fig. 8), which results in a 17% improvement in disturbance rejection and agility performance. In addition to optimizing the control strategy, the computational bottleneck faced by the aircraft during target detection and tracking tasks also needs to be addressed. For example, [216] introduced a heterogeneous parallelization strategy that integrates target detection and tracking algorithms with UAV control software. Then, based on target tracking data, a gain-scheduled PID controller further optimizes flight control to ensure efficient task execution

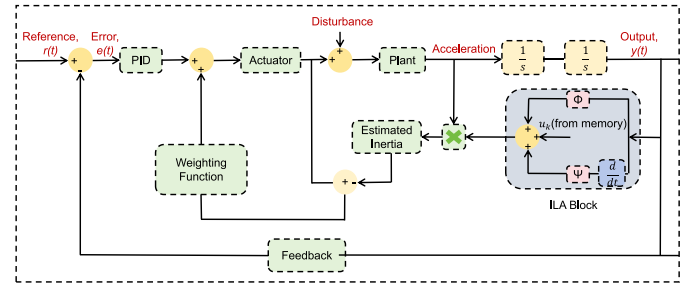


Fig. 8. A schematic diagram of PID-ILAFC [215]. Specifically, the PID controller generates control signals, while the ILA module compensates by storing and learning error information to improve the disturbance rejection and dynamic performance of the system. Moreover, the feedback loop corrects errors to ensure that the output $y(t)$ accurately tracks the reference input $r(t)$.

in complex environments. Therefore, AI-based PID control approaches significantly improve UAV performance in tasks such as disturbance rejection, target tracking, and agility, while serving as a key enabler for enhanced efficiency and robustness within LAE networks.

· **AI-based MPC:** AI can optimize the aircraft model by learning from the historical data, such that improving the accuracy of state predictions and adjusting control decisions in real-time, which enables the aircraft to respond more flexibly and quickly to environmental changes. For example, [217] proposed a UAV control approach that combines neural network learning with NMPC to enhance the tracking accuracy of the aircraft under external disturbances and internal structural changes. Moreover, the authors introduced a novel framework integrating MPC and DRL methods, where the MPC component operates at a lower frequency to provide baseline control inputs, while the DRL component adjusts the outputs generated by MPC at a higher frequency [218]. These highlight that the AI-based MPC has become a critical tool for enhancing aircraft performance, and it is an essential technology in LAE networks. By learning and adapting in real time, AI-based MPC can ensure that the aircraft performs stable and efficient tasks under varying factors such as wind changes, complex terrain, and dynamic obstacles. This makes it suitable for a wide range of applications, including urban air mobility, aerial logistics, agricultural monitoring, and environmental surveillance.

· **HITL-RL-driven Coordination:** HITL-RL is an intelligent training framework that integrates human expert knowledge with RL algorithms. Specifically, HITL-RL relies on three core mechanisms: reward shaping, action injection, and interactive learning [219]. Building on these, HITL-RL provides targeted advantages in safety-critical and multi-agent coordination tasks involving low-altitude autonomous platforms. In particular, the authors demonstrated that incorporating reward shaping in HITL-RL significantly improves training efficiency and task robustness, with TD3-H achieving a success rate of 0.94 in complex urban environments [220]. Moreover, the authors proposed a HITL-DDPG framework for continuous multi-UAV formation control [221], which incorporates an action injection mechanism that allows human operators to override system decisions in high-risk situations, thereby improving the navigation success rate from 0.78 to 0.83 and reducing the

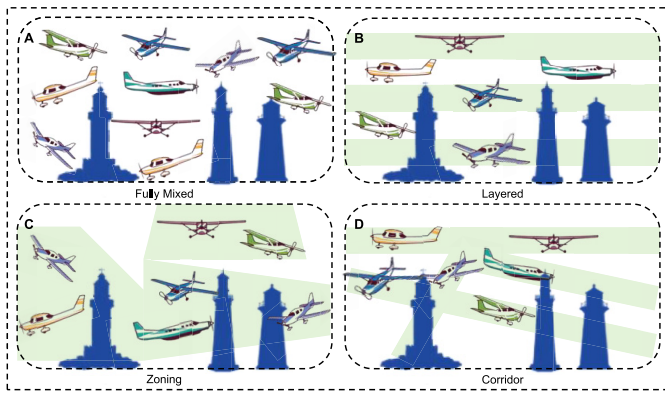


Fig. 9. An illustration of airspace management for low-altitude aircraft. *Part A* shows the fully mixed management, *Part B* presents the layered management, *Part C* depicts the zoning management, and *Part D* demonstrates the corridor management.

weighted average step size from 70.35 to 64.53. Furthermore, the authors introduced an interactive learning mechanism that integrates human demonstrations and trajectory feedback to inform policy updates [222], which enables the system to align more closely with context-specific human intent and improve generalization across heterogeneous coordination environments. In addition, when combined with safe RL [223] and meta-modeling [224] techniques, HITL-RL ensures policy adaptability, as well as accountability, verifiability, and human-compatible behavior under operational uncertainty.

2) *Efficient Airspace Management*: Air traffic management (ATM) focuses on the scheduling and management of the low-altitude airspace to ensure the safe operation of aircraft and the efficient utilization of airspace resources [225]. Several airspace management models have been proposed and studied, as shown in Fig. 9, each with its unique advantages and challenges in terms of flight safety and airspace efficiency.

• **Fully Mixed Management**: The fully mixed management model is an unstructured approach, where the aircraft do not follow the fixed flight paths [226]. Instead, it relies on global coordination and real-time information sharing to maintain safe separation between aircraft. While this model provides flexibility and accommodates various flight demands, the lack of a clear structural framework may introduce complexity in management, thereby reducing the efficiency, particularly in high-demand scenarios where the coordination becomes a limiting factor. In such cases, the model can benefit from real-time data exchange supported by 5G-A networks, dedicated air-to-air communication channels, satellite communication, and ground relay systems, which help resolve conflicts dynamically and enhance overall efficiency.

• **Layered Management**: This divides airspace into multiple altitude layers to reduce conflicts between aircraft. For example, [226] suggested that each altitude layer can be assigned specific flight speeds to improve flight efficiency. The authors in [228] divided the airspace below 400 feet into several layers based on the minimum safety operational diameter. Moreover, a three-layer airspace design was proposed, where each layer group includes both conflict-free and operational layers [229]. However, layered management may be affected by the uneven

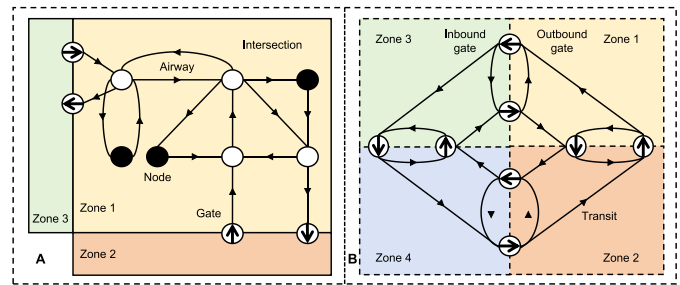


Fig. 10. The network structure based on zones [227]. *Part A* focuses on the structure and flow within a single zone, thereby illustrating how movement occurs through nodes and gates within the zone. *Part B* shows the overall structure across multiple zones, where gates serve as entry and exit points between different zones. Moreover, transit edges represent the movement of UAVs from an inbound gate in one zone to an outbound gate in another.

distribution of airspace resources, which could reduce the overall capacity and lead to relatively poor noise control. To enhance this model, real-time monitoring and dynamic adjustments across layers can be implemented. For instance, cloud-based air traffic management systems can assess traffic conditions as they occur and re-assign aircraft between layers to alleviate congestion and optimize resource allocation.

• **Zoning Management**: The zoning management model divides airspace into multiple independent regions, allowing different flight operations to occur simultaneously. For instance, in [227], the authors partitioned the airspace into several regions, with each region managed by a zone service provider (ZSP). Within each region, individual spaces offer navigation information and flight instructions to ensure safety (Fig. 10). Although this model helps optimize capacity, safety, and efficiency, flying across regions requires complex coordination and may result in higher noise levels. In this context, seamless communication between ZSPs is essential and can be achieved through automated communication protocols (e.g., API-driven service meshes) and edge computing platforms, while AI-enhanced decision-making frameworks streamline cross-zone coordination for uninterrupted transitions.

• **Corridor Management**: The corridor management approach minimizes conflicts and noise in the airspace by providing fixed flight paths for aircraft. The authors designed an airspace system with multiple bidirectional traffic corridors, each separated by a safe distance [230]. Furthermore, the authors introduced a tunnel airspace concept specifically tailored for UAVs, which enables dynamic design and regulates UAV traffic through no-fly zones [231]. However, the drawback of this management model is its lack of flexibility, which limits the adaptability for dynamic flight path adjustments. To overcome this issue, integrating ground-to-air (G2A) communication networks and multi-constellation GNSS navigation systems can enable adaptive corridor reconfiguration based on real-time demand, thus maintaining safety margins during peak operations. In conclusion, different airspace management models have varying applicability depending on the scenario and specific needs. Choosing the appropriate model can significantly enhance the efficiency and safety of low-altitude airspace utilization, which is crucial for the development of the LAE networks.

TABLE VII
SUMMARY OF THE FUSION OF FLIGHT CONTROL WITH AIRSPACE MANAGEMENT TECHNOLOGY

	Technologies	References	Pros & Cons
Advanced-Flight Control	AI-based PID	[214]–[216]	✓Effectively addressing external disturbances and internal uncertainties ✓Improving the accuracy and performance of the flight control system ✕Struggling to maintain stability in rapidly changing and uncertain environments
	AI-based MPC	[217], [218]	✓Considering future states and constraints to improve decision-making ✓Enabling flexible and adaptive control in complex and dynamic environments ✕Involving high computational demands for real-time optimization and state prediction
	HITL-RL-driven Coordination	[219]–[224]	✓Enhancing policy safety and generalization through human-guided training ✓Achieving context-aware multi-agent coordination under uncertainty ✕Requiring expert supervision and additional feedback for effective learning
Efficient-Airspace Management	Fully Mixed Management	[226]	✓Relying on global coordination and real-time information to ensure safe separation ✕Increasing coordination difficulty in high-demand environments
	Layered Management	[226], [228], [229]	✓Reducing aircraft conflicts by dividing airspace into layers ✕Unevenly distributing airspace may reduce capacity and affect noise control efficiency
	Zoning Management	[227]	✓Enabling different flight tasks to proceed simultaneously ✕Requiring complex coordination for inter-regional flights, and increasing noise levels
	Corridor Management	[230], [231]	✓Minimizing conflicts and noise through fixed flight paths ✕Lacking flexibility, which may hinder adaptability in complex flight environments

In summary, the fusion of flight control and airspace management plays a critical role in improving the safety and efficiency of low-altitude airspace operations. While the key technologies required for each have been outlined in Table VII, existing research has yet to explore the comprehensive fusion. In response to this gap, recent studies have shown that integrating GAI with communication, navigation, and flight control empowers UAV systems with more efficient path planning, greater operational safety, and smoother trajectory generation. For instance, generative adversarial imitation learning (GAIL) was applied to virtual navigation training to imitate expert trajectories [232], which improves UAV path planning and generalization in complex 3D environments. Building on this, a multi-agent PPO-based GAIL framework with parallel sampling was proposed, which boosts the target discovery rate to 73.33% in cooperative UAV search tasks, while increasing the average system cost by only 1.11% [233], thus achieving an effective trade-off between efficiency and safety. In addition, the BezierVAE architecture was designed to extend TrajVAE by introducing a Bezier curve output layer [234], which improves trajectory smoothness in both spatial and velocity domains, reduces reconstruction error by 91.3%, and lowers trajectory roughness by 83.4%, thereby demonstrating stronger generalization and greater adaptability to safety verification tasks.

Moreover, leveraging GAI across computing, communication/flight control, and airspace management technologies facilitates robust resource coordination and system stability in LAE environments. For example, [235] introduced a Lyapunov-based mechanism that integrates GDM with RL, so that the system maintains long-term stability and achieves efficient resource allocation under conflicting objectives and external disturbances. Furthermore, in [236], the authors developed a GDM-based multi-agent DDPG algorithm that incorporates the Hungarian algorithm (HG-MADDPG), thus achieving up to 20.35% reduction in average task completion latency under dynamic resource conditions. However, a lack of dynamic network topology consideration may limit the performance of the system in complex environments. In addition, in heterogeneous and service-diverse RIS scenarios, the

authors combined GAI with distributed reinforcement learning (DBRL) to design an on-demand scheduling mechanism [237], which maintains a stable quality of service satisfaction rate (QoSSR) around 1.0 even under limited bandwidth in virtual reality video scenarios. Collectively, these developments underscore the necessity of further exploring GAI-enabled fusion to advance intelligent coordination and operational resilience within LAE networks.

IV. MULTI-TECHNOLOGY INTEGRATION FOR LOW-ALTITUDE APPLICATIONS

In this section, we explore various applications of multi-technology integration in real-world scenarios. Specifically, logistics scenarios (Fig. 11) reflect high-frequency and stability-oriented commercial tasks. Moreover, rescue scenarios (Fig. 12) involve time-sensitive tasks in complex and unstructured environments. Furthermore, transportation scenarios (Fig. 13) address long-term and system-level airspace operations. In addition, inspection scenarios (Fig. 14) focus on tasks related to infrastructure maintenance, industrial surveying, and public safety monitoring. Moreover, we present pilot deployments that correspond to these scenarios.

A. Low-Altitude Logistics

When UAVs perform mid- to long-distance transportation (such as from a warehouse to a retail store), the goal is to ensure reliable and efficient delivery across wide areas. Specifically, during the remote stages of the flight, UAVs may traverse rural, mountainous, or large rivers, where 5G-A signals become weak or unreliable. In such cases, LEO satellite communication ensures seamless connectivity [238]. As the UAVs approach the retail store in an urban area, 5G-A offers high-speed and low-latency communication, which allows the UAVs to quickly respond to ground commands and transmit flight status data in real time [239]. Moreover, LiDAR systems detect distant obstacles and provide geometric mapping under changing terrain and weather conditions [240], [241]. In this setting, GAI improves UAV resilience and autonomy through two key mechanisms. One mechanism

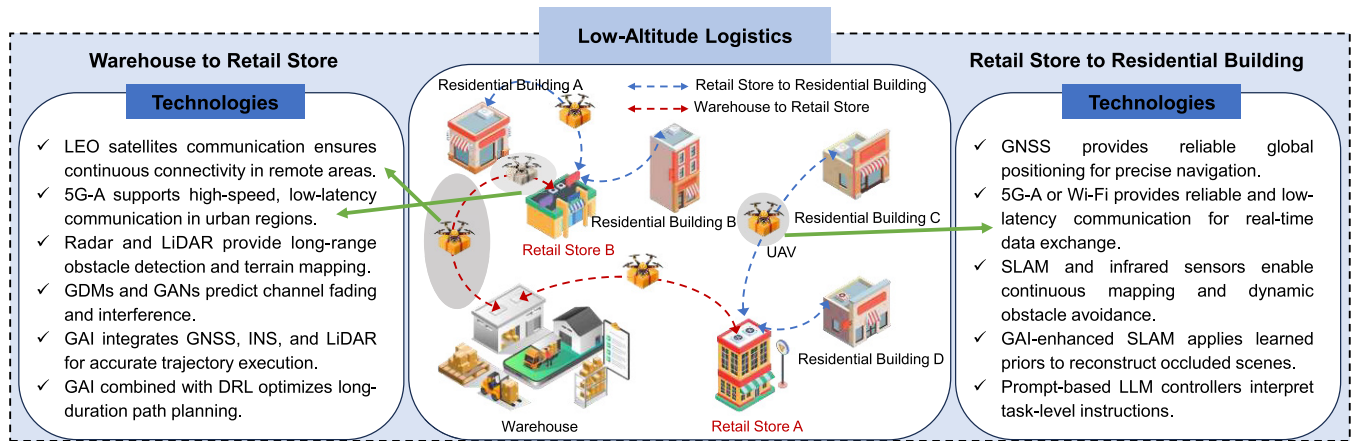


Fig. 11. The role of multi-technology integration in optimizing UAV operations for logistics, which focuses on two key scenarios: warehouse to retail store and retail store to residential building.

leverages GDMs or GANs to predict channel state variations across geographic regions, such as signal fading, blockage, and interference. By learning spatiotemporal patterns in the wireless environment, these models help ensure stable cross-region connectivity and proactive link adaptation. Another mechanism uses GAI-powered sensor fusion and DRL, which combines GNSS, INS, and LiDAR data to support accurate long-range trajectory execution and optimize long-term flight paths [242], thus reducing drift and uncertainty in low-visibility areas.

In short-distance deliveries (such as from a retail store to a residential building), the goal is to quickly and efficiently complete the “last-mile” [243], which ensures that the UAVs reach their destination precisely and safely within a small area. In particular, GNSS signals are typically stable enough to provide global positioning support, and 5G-A or Wi-Fi networks are sufficient to support real-time data transmission and command updates. The focus shifts to more precise environmental awareness and flight control. For instance, in urban or residential areas, UAVs face numerous dynamic obstacles, such as pedestrians, vehicles, and buildings [244]. To avoid collisions, SLAM continuously updates maps using camera data [245], while infrared sensors detect non-visible obstacles in low-light or foggy conditions [246]. In addition, to improve adaptability, GAI-integrated SLAM systems apply learned priors so that occluded scenes can be reconstructed and motion trends of dynamic obstacles can be predicted, which enhances collision avoidance and trajectory planning. Moreover, prompt-based GAI controllers interpret task-level instructions (e.g., “avoid school zone or land on rooftop”) through LLMs, thereby enabling adaptive task adjustment without human reprogramming. This allows the UAVs to update policies based on environmental feedback and semantic constraints in real time.

B. Low-Altitude Rescue

The key to this application lies in rapid response, precise positioning, and environmental awareness, which guarantees the UAVs can quickly reach target areas and execute tasks

in emergency situations. To be specific, 5G-A and LEO satellite communications provide reliable communication links between UAVs and rescue command centers [247]. This ensures that the UAVs can receive instructions and transmit data continuously in complex or remote rescue scenarios. This is crucial for real-time tracking of rescue progress and sharing on-site information, such as video streams and environmental data. Moreover, infrared sensors and LiDAR offer precise environmental perception in low-light or harsh weather conditions, thereby enabling the detection of potential obstacles, assessment of disaster-affected areas, and identification of rescue targets. For example, infrared sensors detect heat sources [248], such as the body temperature of trapped individuals, to locate survivors or hot spots, so that the UAVs can identify targets in low-light or poor-visibility conditions. In addition, LiDAR provides 3D mapping of disaster zones to help rescue teams plan operations [249]. To enhance autonomous perception under such conditions, GAI models are deployed at the edge, where compressed multi-modal networks infer semantic information from limited sensor input. For instance, GDMs can generate structured 3D maps from sparse data, thus improving the identification of viable paths and likely human figures even in partially observed or degraded environments.

Based on this perception layer, the sensed information is further processed through GAI-enabled edge and cloud collaborative computing. More precisely, edge computing quickly analyzes data on the UAVs or nearby devices, allowing real-time adjustments to flight paths, obstacle avoidance, and target localization [250]. Cloud collaborative computing integrates large-scale data and shares it with other devices to optimize task allocation based on proximity, urgency, and forecast weather conditions [251], thereby improving overall operational coordination. In addition, GDM can simulate diverse rescue scenarios, which enables the system to pre-evaluate task strategies and improve the efficiency and robustness of multi-agent coordination. Furthermore, SLAM enables the UAVs to perform precise localization and map updates even in the areas lacking stable GPS signals, such as underground ruins or deep ravines after an earthquake [252]. To guarantee stable and

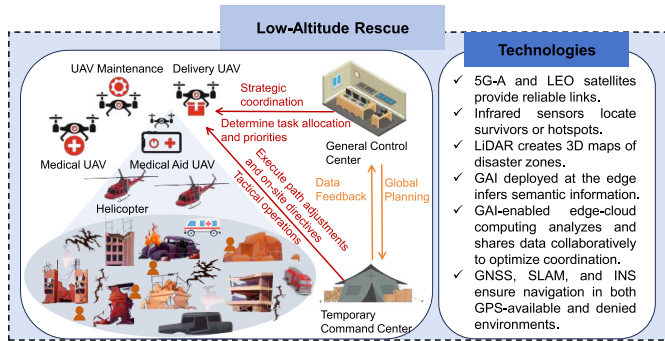


Fig. 12. Leveraging multi-technology integration to enhance UAV efficiency in rescue operations.

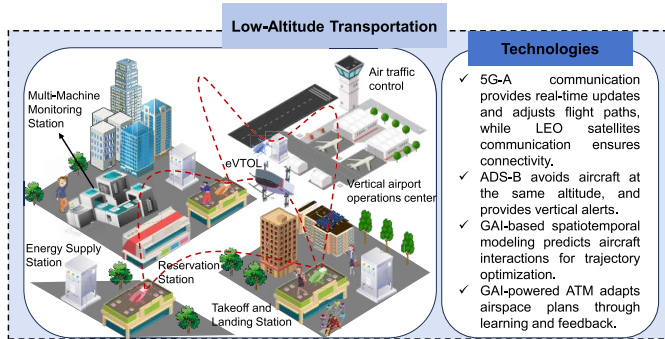


Fig. 13. Enhancing UAV transportation operations through multi-technology integration.

accurate flight paths, GNSS provides broad global positioning support for UAVs in open areas, while INS supplements GNSS when signals weaken or are lost, thereby ensuring continuous navigation for short periods.

C. Low-Altitude Transportation

In this application, ensuring safe interaction between aircraft and airspace coordination is crucial. With its low latency and high bandwidth, 5G-A communication enables aircraft to receive real-time updates on airspace changes, weather conditions, and flight plans, which allows for prompt adjustments to their flight paths. Moreover, LEO satellite communication ensures that the aircraft maintain connectivity with ground control centers, even in remote or signal-restricted areas. At the same time, GAI-driven cloud collaborative computing supports the storage and processing of large-scale real-time data, such as traffic flow, aircraft positions, and airspace dynamics [31], thereby facilitating real-time traffic monitoring, intelligent decision support, and various computational services. Leveraging these capabilities, GAI can help construct a DT of low-altitude airspace, which dynamically simulates collision risks and congestion patterns under dense flight conditions, thus supporting proactive deconfliction and trajectory planning.

Moreover, in congested airspace, ADS-B technology helps aircraft detect and avoid other aircraft at the same altitude, while also providing vertical alerts to prevent potential conflicts. For example, when the trajectories of two aircraft are about to intersect, the ADS-B provides real-time data

on their positions and speeds, thereby enabling aircraft to adjust paths and avoid possible collisions [253]. Therefore, ADS-B compensates for the limitations of visual perception and enhances safety for aircraft. Furthermore, AI-based flight control systems use real-time sensor data and airspace information to predict potential risks and calculate the optimal flight path based on the dynamics of aircraft. To further improve prediction accuracy and adaptability, GAI-based spatiotemporal modeling techniques can be embedded within control modules, which helps anticipate interaction patterns among nearby aircraft and simulate future behavior under uncertain or adversarial conditions, so that proactive trajectory optimization can be achieved in real time. In addition, ATM dynamically adjusts flight plans based on the position, speed, and task requirements of aircraft, which coordinates activities between different aircraft to ensure efficient use of airspace resources and safe separation [254]. Moreover, GAI-powered ATM systems support policy refinement through reward-guided learning and feedback correction, so that the system can update airspace usage plans adaptively in response to evolving constraints or temporary disruptions.

D. Low-Altitude Inspection

In this use case, UAVs are applied to inspect critical infrastructure such as power lines, pipelines, bridges, railways, and industrial facilities [255], [256]. These inspection tasks are often conducted in structurally constrained environments, including narrow access paths, reflective or corroded surfaces, electromagnetic interference, and reduced visibility, all of which pose substantial challenges to conventional sensing and control systems. To address such conditions, UAVs need to perform precise low-altitude navigation near large metallic structures or within confined spaces, so that they can collect ultra-high-resolution images and construct accurate three-dimensional models for comprehensive structural assessment. Moreover, real-time communication is supported by 5G-A or private network slicing, which ensures low-latency transmission of visual and telemetry data to edge or cloud-based platforms. At the same time, radar and LiDAR systems assist in modeling structural surfaces [257], which helps UAVs avoid obstacles and carry out high-fidelity analysis under complex spatial constraints.

Based on this foundation, the introduction of GAI further enhances perception, reasoning, and task adaptability, which significantly improves the performance of UAVs in inspection missions. For example, GDMs reconstruct structural components from sparse or occluded sensor inputs, so that UAVs can operate effectively under limited visibility [258]. Furthermore, vision-language models trained on annotated datasets associate visual features such as cracks, corrosion, and thermal deformation with engineering semantics, which enables automatic labeling and prioritization of structural anomalies. Meta-learning frameworks also support rapid model adaptation to new infrastructure types by using only a few labeled samples, thereby improving generalization in diverse environments. In addition, LLMs interpret high-level task instructions such as “scan the underside of the beam” or “focus on welded joints,”

which allows UAVs to adjust behavior in real time without modifying the underlying code [259]. Moreover, a GAI-integrated DT framework for predictive maintenance combines multi-sensor data and historical patterns, thereby enabling the dynamic simulation of structural degradation trends and the generation of optimized inspection schedules [260]. These capabilities collectively improve the safety, autonomy, and operational efficiency of infrastructure inspection.

E. Pilot Deployments Across Scenarios

Low-altitude aerial mobility is gradually moving beyond conceptual studies, with pilot deployments beginning to emerge in logistics, rescue, transportation, and inspection. Specifically, in logistics, EHang deployed autonomous aerial vehicles in Shenzhen to deliver emergency supplies.⁴ Moreover, in the United States, Zipline received approval from the Federal Aviation Administration (FAA) to conduct routine BVLOS UAV deliveries, which enables scalable medical logistics without the need for visual observers.⁵ For rescue, SoftBank used tethered UAVs to restore communication after the Noto earthquake,⁶ while the Japan UAS Industrial Development Association (JUIDA) coordinated UAV tasks for search and delivery.⁷ In transportation, EHang conducted public demonstration flights of the EH216-S at the urban air mobility (UAM) operation demonstration center in Shenzhen.⁸ Moreover, Joby Aviation and the National Aeronautics and Space Administration (NASA) carried out high-fidelity simulations in the Dallas–Fort Worth area to validate the feasibility of managing more than 100 eVTOL operations per hour using advanced air traffic systems.⁹ In addition, South Korea integrates vertiport design, air corridor planning, and multi-party test flights to support commercial service readiness by 2025.¹⁰ In inspection, JOUAV deployed the CW-15 VTOL UAV with JoLiDAR-1000 to inspect 33.7 km of high-voltage power lines across the karst terrain in Guangxi Province, China.¹¹ In addition, the United Kingdom received one of the first routine BVLOS permissions to autonomously inspect transmission towers using LIDAR and panoramic sensors.¹² Collectively, these deployments reflect the early-stage but expanding application of low-altitude aerial systems in diverse real-world contexts.

F. Key Findings and Insights

Based on the analysis in the preceding subsections, integrated technologies play a vital role in empowering low-altitude aircraft within the LAE networks. In what follows, we outline key areas where further developments are expected to drive critical advancements in operational performance, safety, and scalability.

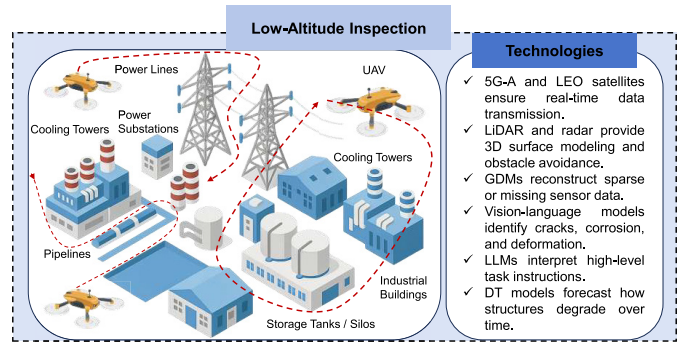


Fig. 14. The role of multi-technology integration in enabling effective UAV operations for inspection.

- *Predictive Interference Resilience*: Low-altitude aircraft often face signal attenuation and interference from overlapping 5G-A, Wi-Fi, and LEO satellite beams, especially near industrial areas. Instead of reactive adjustments, GDM- or GAN-based models can proactively generate spatiotemporal interference maps to guide real-time tuning of frequency bands, antenna modes, and task scheduling [261]. Moreover, federated learning enables shared map construction across aircraft clusters, thus enabling channel adaptation and system resilience [262].
- *Semantic Sensing Recovery*: Although high-precision sensors such as LiDAR and infrared enhance local awareness, their performance can sharply degrade in dense fog, rubble, or underground spaces. In such cases, GAI can go beyond detection and infer environments via weak signal analysis with semantic mapping [263]. For example, using shadow fragments, edge cues, and motion patterns, these models can estimate obstacles and reconstruct spatial geometry for safe navigation. However, such systems require simulated training and continuous adaptation to stay reliable under degraded sensing.
- *Elastic Task Reconfiguration*: A core capability of low-altitude systems lies in their ability to respond to changing environments by promptly reshaping task sequences. In multi-task scenarios, aircraft must support task interruption and rapid switching (e.g., from logistics to emergency). To enable this, GAI-driven arbitration engines continuously assess task value based on urgency, location, and semantic input [236]. When disruptions occur, they can recompose behavior pipelines by assembling modules from a dynamic skill library. Intent messages are then shared with nearby units to support decentralized cooperation and adaptive task handoff.
- *Adaptive Degradation Autonomy*: In extreme flight scenarios such as electromagnetic interference, GPS failure, or hardware malfunction, low-altitude aircraft are vulnerable to cascading faults. To ensure survivability and task continuity, a degradation-aware execution model should be adopted [264]. This model classifies task layers into fallback stages, such as full autonomy, degraded autonomy, and minimal safeguard modes. Moreover, aircraft should integrate self-assessment mechanisms and adaptive mode-switching logic to gracefully transition into

⁴<https://www.ehang.com/news/987.html?>

⁵Zipline BVLOS FAA Approval/.

⁶https://www.softbank.jp/en/sbnews/entry/20240111_01

⁷DroneLife report on JUIDA post-earthquake operations.

⁸EHang press release on Shenzhen UAM demonstration.

⁹Joby & NASA UAM simulation press release.

¹⁰Uncrewed Airspace: Korea UAM Grand Challenge.

¹¹JOUAV official website (Chinese).

¹²Sees.ai announcement on UK BVLOS inspection.

controllable states under partial failure, thereby preserving minimal functions for critical tasks [265].

V. FUTURE DIRECTIONS

In this section, we will present important future directions for collaborative technologies in LAE networks.

A. Intelligent and Adaptive Optimization of Dynamic Airspace

The coexistence of high-density, multi-type aircraft in low-altitude airspace requires overcoming traditional air traffic control constraints by establishing an intelligent and adaptive management system. By integrating GAI-enabled DRL [266], game theory [267], and DT systems [134] alongside various low-altitude sensors and communication devices, the system can predict environmental changes, model task dynamics, and optimize control strategies. In addition, establishing cross-domain transfer learning mechanisms would enable the efficient reuse of GDM-generated synthetic DT data across heterogeneous low-altitude aircraft platforms, thus reducing reliance on real-world samples while enhancing model generalization and deployment efficiency. Moreover, airspace usage rights can be managed through an auction model [268] that incorporates different time slots and priority tasks to design differentiated pricing strategies. Furthermore, blockchain smart contracts [269] can then be used to automatically execute transactions, thus incentivizing users to actively avoid peak airspace and optimizing flight management within the LAE networks.

B. Security and Privacy Protection of LAE Networks

The openness and data mobility of LAE networks expose them to multiple security threats, which necessitate a multi-layered defense system. For example, the cloud-edge-end collaborative security architecture can protect data confidentiality and integrity by deploying trusted execution environments (TEEs) on aircraft [270], running intrusion detection systems (IDS) at edge nodes [271], and utilizing homomorphic encryption at the cloud level [272]. In terms of the privacy protection, LAE networks involve substantial amounts of user data (e.g., flight trajectories and personal identities). Therefore, techniques such as differential privacy [273] and federated learning [274] can anonymize and protect user data, thus ensuring that the privacy is not compromised during data sharing and analysis. To further enhance the security and privacy, network resilience is required to handle unexpected events such as cyberattacks or equipment failures. Thus, AI-based anomaly detection [275], autonomous reconfiguration mechanisms [276], and multi-path communication technologies [277] can be employed to improve system adaptability and recovery capabilities, so that continuous protection of critical data and services is ensured.

C. Sustainable Power and Energy Management of Aircraft

The development of LAE networks may face challenges in sustainable energy supply, particularly with the large-scale deployment of high-energy-consuming aircraft such as

eVTOLs and UAVs, which require the enhanced dynamic response and sustainability in energy systems. Future research can focus on aerial energy relay networks [278], where solar-powered UAVs function as the charging stations to create an air-to-air energy supply chain. Furthermore, bio-hybrid energy systems could be explored [279], which use microbial fuel cells along with rainwater and carbon dioxide collected from aircraft surfaces to generate power. MmWave beamforming and laser energy transmission [280] could enable wireless charging of aircraft during hovering or low-speed cruising. In addition, energy consumption prediction models that couple flight tasks, weather conditions, and battery health status could be developed, with GAI-enabled DRL used to optimize power distribution [281].

D. Quantum-Driven Intelligence Coordination for Airspace

As LAE networks evolve toward massive-scale aircraft coordination in real time, classical computational paradigms may struggle to handle the exponential growth of state-action spaces and multi-agent constraints. To address this, integrating quantum computing, particularly quantum annealing [282] and variational quantum circuits [283], into airspace management makes it possible to accelerate route optimization, task allocation, and conflict resolution in high-density low-altitude environments. In particular, quantum-inspired algorithms can be embedded into DT platforms to simulate thousands of parallel flight scenarios under uncertainty [284]. Moreover, hybrid classical-quantum models can support the optimization of secure and ultra-fast communication protocols by leveraging quantum entanglement properties [285], which in turn strengthens the resilience of collaborative decision-making among aircraft clusters in contested or congested airspace.

E. Self-Evolving Cognitive Agents for Generative Governance

Future research should advance self-evolving cognitive agents for low-altitude airspace with generative and interactive capabilities to enhance autonomy, adaptability, and safety. One promising direction is the fusion of GAI and XR technologies, which can enable cross-modal semantic alignment of visual, linguistic, and spatial data in XR environments to enhance task comprehension and command execution [142]. In parallel, the development of lightweight GAI models optimized for resource-constrained XR-aircraft platforms is essential to ensure real-time performance and operational stability under real-world conditions [286]. Another critical avenue is the incorporation of HITL-RL into multi-agent systems, which has the potential to support preference-driven policy adjustment, task reconfiguration, and dynamic risk intervention [219]. By leveraging human feedback to perform localized policy corrections and knowledge transfer in scenarios with unstable communication or sudden task changes, the system can enable a self-evolving airspace architecture characterized by continuous learning, human supervision, and safety assurance.

F. LAE-LEO Collaboration for 3D Airspace Coverage

As low-altitude airspace tasks grow and coverage expands, the integration of LAE networks with LEO satellite networks

will be essential for achieving seamless 3D coverage. Specifically, the dynamic orbits, low latency, and wide coverage of LEO satellites facilitate reliable cross-regional communication for LAE systems, particularly in areas with limited ground infrastructure or under extreme environmental conditions [287]. To support dynamic cross-domain collaboration, future research could focus on developing an intelligent and cooperative spectrum management architecture. This architecture integrates GAI-based spectrum prediction [288] and interference modeling [289] to identify heterogeneous and multi-source interference in real time, so that spectrum reconfiguration and channel scheduling can be conducted based on the status of uplink and downlink channels. Furthermore, a unified air-ground-space DT system can be established to realize resource virtualization and link optimization between aircraft and LEO satellites [290]. Such a system will facilitate the construction of a highly reliable and reconfigurable 3D collaborative network, thereby enabling task-level self-organization and intelligent interconnection in next-generation airspace operations.

VI. CONCLUSION

In this article, we conducted a comprehensive analysis of specific aspects of LAE networks. We began by describing the distinct advantages of LAE networks over traditional UAV networks and introduced the relevant standards and core architecture that support their development. Next, we investigated the integration of key technologies within LAE networks, including the integration of communication, sensing, and intelligent computing, collaboration of positioning, navigation, and surveillance, and the fusion of flight control with airspace management. We also highlighted the potential of GAI in enhancing the effectiveness of these technologies in LAE networks. Following this, we demonstrated that how these integrated technologies enable the sustainable operation of real-world LAE applications. Finally, we outlined the future research directions and highlighted the potential of collaborative technologies in LAE networks for intelligent and adaptive optimization, security and privacy protection, sustainable energy and power management, quantum-driven coordination, generative governance, and 3D airspace coverage.

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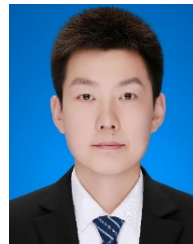
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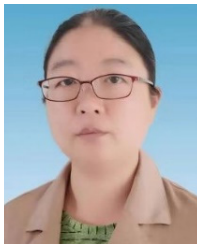
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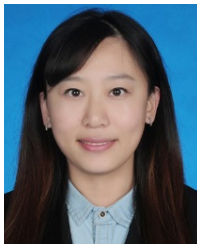
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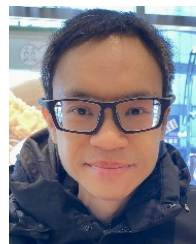
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